

# Peter Richtárik: Curriculum Vitae

## 1. CONTACT DETAILS

Address: Office 3145, Building 12, 4700 KAUST, Thuwal 23955-6900, Kingdom of Saudi Arabia  
Email: [peter.richtarik@kaust.edu.sa](mailto:peter.richtarik@kaust.edu.sa)  
Website: <https://richtarik.org>

## 2. RESEARCHER IDs

dblp: <https://dblp.org/pid/62/8001.html>  
orcid: <https://orcid.org/0000-0003-4380-5848>  
Web of Science Researcher ID: O-5797-2018

## 3. RESEARCH INTERESTS

- ◇ machine learning, federated learning, empirical risk minimization
- ◇ big data optimization, convex and non-convex optimization; 0th, 1st, and 2nd order optimization methods
- ◇ randomized algorithms, randomized coordinate descent, stochastic gradient descent, variance reduction
- ◇ randomized numerical linear algebra
- ◇ parallel and distributed computing, supercomputing, gradient compression

## 4. ACADEMIC POSITIONS

2019–now **Professor**, Computer Science, King Abdullah University of Science and Technology (KAUST), Kingdom of Saudi Arabia  
2022–2023 **Adjunct Professor**<sup>1</sup>, Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, United Arab Emirates  
2017–2019 **Visiting Professor**, Moscow Institute of Physics and Technology, Russia  
2017–2019 **Associate Professor**, Computer Science, KAUST, Kingdom of Saudi Arabia  
2016–2019 **Associate Professor (Reader)**, Mathematics, University of Edinburgh  
2013 **Invited Visiting Scientist**, Simons Institute for the Theory of Computing, UC Berkeley  
2009–2016 **Assistant Professor (Lecturer)**, School of Mathematics, University of Edinburgh  
2007–2009 **Postdoctoral Fellow**, Center for Operations Research and Econometrics and Department of Mathematical Engineering, Catholic University of Louvain, Belgium (host: Yu. Nesterov)

## 5. EDUCATION

2007 **PhD, Operations Research, Cornell University**  
2006 MS, Operations Research, Cornell University  
2001 Mgr, Mathematics, Comenius University, Slovakia, 100% academic grades, ranked #1  
2001 Bc, Management, Comenius University, Slovakia, 100% academic grades, ranked #1  
2000 Bc, Mathematics, Comenius University, Slovakia, 100% academic grades, ranked #1

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<sup>1</sup>This position allowed me to supervise Dr. Sarit Khirirat – a postdoc located at MBZUAI.

## 6. AWARDS & RECOGNITIONS

Awards explicitly addressed to my students, postdocs or coauthors for a talk, poster or paper based on joint research with me are listed in the section “8.4 My Team: Awards and Recognitions”.

|      |                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------|
| 2026 | <b>2023 Charles Broyden Prize<sup>2</sup></b> (for paper [94])                                                                   |
| 2025 | <b>Spotlight Paper at ICLR 2025</b> (for paper [244])                                                                            |
| 2025 | <b>AMA Distinguished Speaker</b> , The Hong Kong Polytechnic University                                                          |
| 2024 | <b>Top 8–14 Author at NeurIPS 2024</b> in terms of the number of papers accepted to the conference <sup>3</sup>                  |
| 2024 | <b>2022 Charles Broyden Prize<sup>4</sup></b> (for paper [139])                                                                  |
| 2024 | <b>Oral Paper at NeurIPS 2024</b> (0.4% acceptance rate <sup>5</sup> ; paper [249])                                              |
| 2024 | <b>Spotlight Paper at NeurIPS 2024</b> (for paper [243])                                                                         |
| 2024 | <b>Oral Paper at ICML 2024</b> (1.5% acceptance rate; paper [231])                                                               |
| 2023 | <b>#1 in Asia and #3 in Europe in Machine Learning according to CSRankings.org<sup>6</sup></b>                                   |
| 2023 | Research.com Computer Science in Saudi Arabia Leader Award                                                                       |
| 2023 | Research.com Mathematics in Saudi Arabia Leader Award                                                                            |
| 2023 | <b>Oral Paper at ICLR 2023</b> (for paper [184])                                                                                 |
| 2022 | <b>Top 10–14 Author at NeurIPS 2022</b> in terms of the number of papers accepted to the conference <sup>7</sup>                 |
| 2022 | Research.com Rising Star of Science <sup>8</sup> , global rank 214 among all fields of science                                   |
| 2022 | <b>Spotlight Paper at ICLR 2022</b> (for paper [156])                                                                            |
| 2021 | <b>Oral Paper at NeurIPS 2021</b> (less than 1% acceptance rate; paper [167])                                                    |
| 2021 | <b>2020 COAP Best Paper Award<sup>9</sup></b> (for paper [65])                                                                   |
| 2021 | <b>One of the 10 Most Cited Articles Published in SIMAX Since 2019<sup>10</sup></b> for paper [56]                               |
| 2020 | <b>Best Paper Award at the NeurIPS 2020 Workshop on Scalability, Privacy, and Security in Federated Learning</b> for paper [135] |
| 2020 | <b>Top 30–50 author at ICML 2020</b> (in number of papers accepted)                                                              |
| 2020 | 1st Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [39]                                        |
| 2020 | 3rd Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [44]                                        |
| 2020 | 3rd Most Downloaded Paper in “SIAM J. on Optimization” for paper [57]                                                            |
| 2020 | 4th Most Downloaded Paper in “SIAM J. on Optimization” for paper [21]                                                            |
| 2019 | 1st Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [39]                                        |
| 2019 | 4th Most Downloaded Paper in “SIAM J. on Optimization” for paper [57]                                                            |
| 2019 | 5th Most Downloaded Paper in “SIAM J. on Optimization” for paper [21]                                                            |
| 2019 | 10th Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [44]                                       |
| 2019 | <b>Interviewed by Robin.ly for their “Leaders in AI” platform at NeurIPS 2019<sup>11</sup></b>                                   |

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<sup>2</sup>The prize was announced in 2026.

<sup>3</sup><https://papercopilot.com/paper-list/neurips-paper-list/neurips-2024-paper-list/>

<sup>4</sup>The prize was announced in 2024.

<sup>5</sup>61 papers out of more than 15,000.

<sup>6</sup>As all rankings, these rankings have to be taken with a huge grain of salt. According to CS Rankings in the 5 year period 2019–2023, I rank #1 in Machine Learning in all of Asia: 1) Peter Richtárik at KAUST (14), 2) Sung Ju Hwang at KAIST (13.6), 3) Jinwoo Shin at KAIST (13.2), 4) Jun Zhu at Tsinghua (12.3), 5) Masashi Sugiyama at Tokyo (12), 6) Junchi Yan at Shanghai Jiao Tong (10.4) 7) Taiji Suzuki at Tokyo (9.6), 8) Kun Zhang at MBZUAI (9.3). If KAUST was in Europe, I would rank #3 in Europe in the same metric: 1) Andreas Krause at ETH (17.5), 2) Bernhard Schölkopf at Max Planck (14.9), 3) Peter Richtárik at KAUST (14), 4) Volkan Cevher at EPFL (12.5), 5) Max Welling at Amsterdam (12.2), 6) Stephan Günnemann at TU Munich (12.1), 7) Yishay Mansour at Tel Aviv (9.6), 8) Shie Mannor at Technion (9.3), 9) Francis Bach at ENS (9.2), 10) Shimon Whiteson at Oxford (8.9), 11) José Miguel Hernández-Lobato at Cambridge (8.8).

<sup>7</sup><https://papercopilot.com/paper-list/neurips-paper-list/neurips-2022-paper-list/> (my team had 12 papers accepted; I was a coauthor on 9)

<sup>8</sup><https://research.com/u/peter-richtarik>

<sup>9</sup>For the best paper published in Computational Optimization and Applications in 2020.

<sup>10</sup><https://sinews.siam.org/Details-Page/10-most-highly-cited-articles-from-simax-since-2019-1>

<sup>11</sup>From Robin.ly LinkedIn Post: “We are interviewing the world’s leading AI academics this week at NeurIPS2019. Look forward to sharing much more on the state of AI research, how it’s fueling AI commercialization & what we can expect from AI

|             |                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------|
| 2019        | <b>Best NeurIPS Reviewer Award</b> <sup>12</sup>                                                |
| 2019        | <b>Distinguished Speaker Award</b> , Int. Conf. on Continuous Optimization, Berlin, Germany     |
| 2018        | <b>Best NeurIPS Reviewer Award</b> <sup>13</sup>                                                |
| 2018        | 2nd Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [39]       |
| 2018        | 6th Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [44]       |
| 2017        | 1st Most Read Paper in “Optimization Methods and Software” for paper [41]                       |
| 2017        | 1st Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [39]       |
| 2017        | 1st Most Trending Paper in “Mathematical Programming” for paper [10]                            |
| 2017        | <b>Announcement of “Federated Learning”</b> by Google (based on papers [51, 52])                |
| 2016–2017   | 2nd Most Downloaded Paper in “SIAM J. on Optimization” for paper [21]                           |
| 2016        | <b>SIAM SIGEST Outstanding Paper Award</b> for paper [21]                                       |
| 2016        | <b>EUSA Best Research or Dissertation Supervisor Award</b> <sup>14</sup> (2nd Prize)            |
| 2016–now    | <b>Turing Fellow, The Alan Turing Institute</b> , London                                        |
| 2016        | <b>EPSRC Fellowship in Mathematical Sciences</b> <sup>15</sup>                                  |
| 2014        | Nominated for the <b>Chancellor’s Rising Star Award</b> <sup>16</sup> , University of Edinburgh |
| 2013        | <b>Simons Institute Visiting Scientist Fellowship</b> , UC Berkeley                             |
| 2013        | Nominated for the <b>2014 Microsoft Research Faculty Fellowship</b> <sup>17</sup>               |
| 2011 & 2012 | Nominated for the <b>Innovative Teaching Award</b> , University of Edinburgh                    |
| 2011–2017   | <b>Honorary Fellow</b> , Heriot-Watt University                                                 |
| 2007        | <b>CORE Fellowship</b> , Université catholique de Louvain                                       |
| 2002        | <b>Cornell University Graduate Fellowship</b>                                                   |
| 2001        | <b>Dean’s Prize and Rector’s Prize</b> , Comenius University                                    |
| 1992–2001   | Winner of Numerous Mathematical Olympiads and Competitions                                      |

## 7. GRANTS

### 7.1 MY GRANTS<sup>18</sup>

|           |                                                                                                                                                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026–2028 | <b>\$410,413 (PI)</b> , Industrial Research Grant from Saudi company Tahakom, “Optimizing LLM Efficiency through Precision and Parameter Reduction”, Joint with: Marco Canini (KAUST, PI) and Suhaib Fahmy (KAUST, PI)                            |
| 2026–2027 | <b>\$100,000 (PI)</b> , KAUST Small Translational Grant (sTRG), “TRACE: Transforming Textbooks into Verified, Interactive Courses using Retrieval-Augmented Generation”, Joint with: Laurent Condat (research scientist), Ivan Ilin (PhD student) |
| 2026–2027 | <b>\$650,000 (PI)</b> , KAUST Baseline Research Grant <sup>19</sup>                                                                                                                                                                               |
| 2025–2027 | <b>\$100,000</b> , Google PhD Fellowship in Algorithms & Optimization for Kaja Gruntkowska <sup>20</sup>                                                                                                                                          |
| 2025–2028 | <b>\$675,000 (PI)</b> , Competitive Research Grant (CRG)                                                                                                                                                                                          |
| 2025–2028 | <b>1,600,000 SAR (PI)</b> , RDIA Baseload Applied Grant (BAG)                                                                                                                                                                                     |
| 2025–2026 | <b>\$650,000 (PI)</b> , KAUST Baseline Research Grant                                                                                                                                                                                             |

in the next decade. Spotlight interviews with Yoshua Bengio, Peter Richtárik, Charles Onu, Max Welling, Shimon Whiteson, Sharon Zhou, Liwei Wang, Song Han & many more. ”

<sup>12</sup>“Thank you for all your hard work reviewing for NeurIPS 2019! We are delighted to inform you that you were one of the 400 highest-scoring reviewers this year! You will therefore be given access (for a limited period of time) to one free registration to this year’s conference; you will later receive additional information by email explaining how to access your registration. ”

<sup>13</sup>“We are delighted to inform you that you were one of the 218 highest-scoring reviewers this year! You will therefore be given access (for a limited period of time) to one free registration to this year’s conference.”

<sup>14</sup>EUSA = Edinburgh University Students’ Association. One first and one second prize are given each year across all disciplines and levels of seniority at the University of Edinburgh.

<sup>15</sup>In total, 5 fellowships in mathematics were awarded in the UK in this round at all levels of seniority.

<sup>16</sup>One of two nominated from the School of Mathematics.

<sup>17</sup>Selected universities can nominate a single candidate. No European scientists got the award in 2014.

<sup>18</sup>All small grants (value below \$10k) are excluded from this list. The total value of the 16 small grants excluded is £42,090. Funding from the VCC and ECRC centers at KAUST is excluded from this list.

<sup>19</sup>Unrestricted basic research funding offered each year to KAUST Professors.

<sup>20</sup>An unrestricted gift of \$50,000 per year for two years

|           |                                                                                                                                                                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024–2029 | <b>\$11,000,000 (Co-PI)</b> , KAUST Center of Excellence for Generative AI                                                                                                                                                                                                                                        |
| 2024–2025 | <b>\$650,000 (PI)</b> , KAUST Baseline Research Grant                                                                                                                                                                                                                                                             |
| 2024      | <b>1,000,000 SAR (PI)</b> , SDAIA-KAUST Center of Excellence in Data Science and Artificial Intelligence                                                                                                                                                                                                          |
| 2023–2024 | <b>\$60,000 (PI)</b> , SDAIA-KAUST Center of Excellence in Data Science and Artificial Intelligence                                                                                                                                                                                                               |
| 2023–2024 | <b>\$650,000 (PI)</b> , KAUST Baseline Research Grant                                                                                                                                                                                                                                                             |
| 2022–2023 | <b>\$60,000 (PI)</b> , SDAIA-KAUST Center of Excellence in Data Science and Artificial Intelligence                                                                                                                                                                                                               |
| 2022–2023 | <b>\$40,000 (PI)</b> , Top-up to KAUST Baseline Research Grant                                                                                                                                                                                                                                                    |
| 2022–2023 | <b>\$540,000 (PI)</b> , KAUST Baseline Research Grant                                                                                                                                                                                                                                                             |
| 2021–2022 | <b>\$540,000 (PI)</b> , KAUST Baseline Research Grant                                                                                                                                                                                                                                                             |
| 2021–2022 | <b>\$100,000 (PI)</b> , AI Initiative Funding                                                                                                                                                                                                                                                                     |
| 2020–2021 | <b>\$540,000 (PI)</b> , KAUST Baseline Research Grant                                                                                                                                                                                                                                                             |
| 2020      | <b>\$100,000 (PI)</b> , AI Initiative Seed Funding, “Algorithmic, Systems and Privacy Aspects of Split Learning”, Joint with: Marco Canini (KAUST, Co-I) and Panos Kalnis (KAUST, Co-I)                                                                                                                           |
| 2019–2020 | <b>\$200,000 (PI)</b> , Extreme Computing Research Center (ECRC) funding, KAUST, “Optimization for Machine Learning”, Joint with: Tong Zhang (HKUST, PI)                                                                                                                                                          |
| 2019–2020 | <b>\$540,000 (PI)</b> , KAUST Baseline Research Grant                                                                                                                                                                                                                                                             |
| 2018–2019 | <b>£216,843 (Co-I)</b> , Innovate UK Grant, “Renewable Energy Performance Score (REPScore)”, Joint with: Enian (PI), Daniel Friedrich (Edinburgh, PI)                                                                                                                                                             |
| 2018–2021 | <b>\$974,789 (Co-I)</b> , CRG2017 Grant, “Analyzing Large Scale 3D Shape Collections”, Joint with: Peter Wonka (KAUST, PI), Maks Ovsjanikov (École Polytechnique, Co-I)                                                                                                                                           |
| 2017–2019 | <b>RUB 7,960,000 (PI)</b> , Visiting Professor Grant, Moscow Institute of Physics and Technology                                                                                                                                                                                                                  |
| 2018      | <b>\$10,000 (PI)</b> , KICP grant in support of KAUST Research Workshop on Optimization and Big Data, 2018                                                                                                                                                                                                        |
| 2018–2019 | <b>\$400,000 (PI)</b> , KAUST Baseline Research Grant <sup>21</sup>                                                                                                                                                                                                                                               |
| 2017–2018 | <b>\$79,281 (PI)</b> , KAUST Office of Sponsored Research Conference Support Grant URF/1/3347-01, “Optimization and Big Data”, Joint with: Marco Canini (KAUST, PI)                                                                                                                                               |
| 2016–2020 | <b>£70,000 EPSRC CASE<sup>22</sup> PhD Studentship</b> for Filip Hanzely                                                                                                                                                                                                                                          |
| 2017–2018 | <b>\$400,000 (PI)</b> , KAUST Baseline Research Grant                                                                                                                                                                                                                                                             |
| 2016–2017 | <b>\$133,333 (PI)</b> , KAUST Baseline Research Grant (4 months of cover: March-June 2017)                                                                                                                                                                                                                        |
| 2016–2020 | <b>£45,000 (PI)</b> , Amazon Research Grant                                                                                                                                                                                                                                                                       |
| 2016–2020 | <b>£823,211 (PI)</b> , EPSRC Early Career Fellowship in Mathematical Sciences EP/N005538/1, “Randomized Algorithms for Extreme Convex Optimization”                                                                                                                                                               |
| 2016–2020 | <b>\$20,000 (PI)</b> , Amazon EC2 Grant (partner funding associated with the EPSRC Fellowship)                                                                                                                                                                                                                    |
| 2015      | <b>£20,000 (PI)</b> , Alan Turing Institute Scoping Workshop Grant, “Distributed Machine Learning and Optimization”, Joint with: Artur Czuma (Warwick, PI), Ilias Diakonikolas (Edinburgh, PI), Mark Girolami (Warwick, PI), Raphael Hauser (Oxford, PI), John Shawe-Taylor (UCL, PI)                             |
| 2015      | <b>£12,000 (PI)</b> , Alan Turing Institute Scoping Workshop Grant, “Theoretical and Computational Approaches to Large Scale Inverse Problems”, Joint with: Simon Arridge (UCL, PI), John Aston (Cambridge, PI), Carola-Bibiane Schönlieb (Cambridge, PI), Andrew Stuart (Warwick, PI), Jared Tanner (Oxford, PI) |
| 2014–2017 | <b>\$180,000</b> , Google Europe Doctoral Fellowship for Jakub Konečný                                                                                                                                                                                                                                            |
| 2013–2015 | <b>£125,849 (PI)</b> , EPSRC First Grant EP/K02325X/1, “Accelerated Coordinate Descent Methods for Big Data Optimization”                                                                                                                                                                                         |
| 2014–2015 | <b>£40,000 (PI)</b> , School of Mathematics Grant, “Accelerated Coordinate Descent Methods for Big Data Optimization”, matching funding for my postdoc Z. Qu                                                                                                                                                      |
| 2013      | <b>£18,785 (PI)</b> , NAIS Travel Grant, my 2 students spending semester at Berkeley                                                                                                                                                                                                                              |
| 2012–2014 | <b>£66,300 (PI)</b> , NAIS Lecturer Grant, paying for a proportion of my time                                                                                                                                                                                                                                     |
| 2012–2014 | <b>£10,000 (PI)</b> , NAIS Startup Grant                                                                                                                                                                                                                                                                          |

<sup>21</sup>Unrestricted basic research funding offered each year to KAUST Associate Professors.

<sup>22</sup>CASE = Cooperative Awards in Science and Engineering

|           |                                                                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2012–2013 | <b>£49,518 (Co-I)</b> , EPSRC grant EP/J020567/1, “Algorithms for Data Simplicity”, Joint with: Jared Tanner (Oxford, PI)                                                        |
| 2011–2014 | <b>£646,264 (Co-I)</b> , EPSRC and RCUK grant EP/I017127/1, “Mathematics for Vast Digital Resources”, Joint with: Burak Büke (Edinburgh, Co-I) and Jacek Gondzio (Edinburgh, PI) |

## 7.2 GRANTS I HELPED TO PREPARE<sup>23</sup>

|           |                                                                                                                                                                                                |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2014–2019 | £42 million + £5 million, “The Alan Turing Institute”. I am one of a small number of people who helped to prepare Edinburgh’s bid.                                                             |
| 2014–2023 | £4.5 million, EPSRC grant, “Maxwell Institute Graduate School in Mathematical Analysis and Applications”, PI: Anthony Carbery (Edinburgh). I am one of the named PhD supervisors on the grant. |
| 2014–2021 | £5.03 million, EPSRC grant, “Centre for Doctoral Training in Data Science”, PI: Chris Williams (Edinburgh). I am one of 45 named potential PhD advisors at U of Edinburgh.                     |

## 8. MY TEAM

### 8.1 MY TEAM @ KAUST

|                 |                                                                           |
|-----------------|---------------------------------------------------------------------------|
| 08/2026–now     | <b>MS student: Tigran Melkonyan (from AUA, Armenia)</b>                   |
| 07/2026–now     | <b>Research scientist: Xun Qian (from Shanghai, China)</b>                |
| 07/2026–now     | <b>Postdoc: Egor Shulgin (continuing after PhD at KAUST)</b>              |
| 06/2026–now     | Intern: Jan Niklas Kolzenburg (from Frankfurt, Germany)                   |
| 06/2026–now     | <b>Postdoc: Grigory Malinovsky (from MIPT, Russia)</b>                    |
| 12/2025–05/2026 | Postdoc: Artavazd Maranjyan (from KAUST, Saudi Arabia) <sup>24</sup>      |
| 08/2025–now     | <b>MS student: Alina Abdikarimova (from Nazarbayev U, Kazakhstan)</b>     |
| 08/2025–now     | <b>MS student: Alexander Gaponov (from HSE, Russia)</b>                   |
| 08/2025–now     | <b>MS student: Zhirayr Tovmasyan (from Yerevan State U, Armenia)</b>      |
| 08/2025–now     | <b>PhD student: Ammar Mahran (from U of Oxford, UK)</b>                   |
| 06/2025–now     | <b>PhD student: Yassine Maziane (from University of Liege, Belgium)</b>   |
| 05/2025–08/2025 | Intern: Alexander Gaponov (from HSE, Russia)                              |
| 07/2025–05/2026 | Intern: Igor Klimczak (from U of Oxford, UK)                              |
| 05/2025–09/2025 | Intern: Hussein Rammal (from École Polytechnique, France)                 |
| 04/2025–09/2025 | Intern: Adrien Fradin (from École Polytechnique, France)                  |
| 01/2025–now     | <b>PhD student: Ivan Ilin (from Novosibirsk State University, Russia)</b> |
| 10/2024–04/2025 | Intern: Philip Zmushko (from MIPT, Russia)                                |
| 09/2024–07/2025 | Intern: Zhirayr Tovmasyan (from Yerevan State University, Armenia)        |
| 06/2024–11/2024 | Intern: Aadi Rane (from UC Berkeley, USA)                                 |
| 05/2024–09/2024 | Intern: Wojciech Anyszka (from U of Groningen, Netherlands)               |
| 02/2024–05/2026 | Postdoc: Sarit Khirirat (from KTH, Sweden)                                |
| 02/2024–05/2024 | Intern: Simone Maria Giancola (from Bocconi, Italy)                       |
| 01/2024–05/2024 | Intern: Dymitr Lubczyk (from Amsterdam, Netherlands)                      |
| 01/2024–04/2024 | Intern: Kirill Acharya (from MIPT, Russia)                                |
| 01/2024–05/2024 | Intern: Robin Yadav (from UBC, Canada)                                    |
| 01/2024–now     | <b>PhD student: Kaja Gruntkowska (from U of Oxford, UK)</b>               |
| 12/2023–03/2024 | Intern: Anh-Duc Nguyen (from NUS, Singapore)                              |
| 10/2023–03/2024 | Intern: Timur Kharisov (from MIPT, Russia)                                |
| 10/2023–04/2024 | Intern: Georg Meinhardt (from University of Oxford, UK)                   |
| 09/2023–01/2024 | Intern: Marta Pozzi (from Pavia, Italy)                                   |

<sup>23</sup>Large grants which I helped to prepare but where I am not formally an investigator.

<sup>24</sup>next position: Postdoc, EPFL, Switzerland

|                               |                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------|
| 09/2023–02/2024               | Intern: Ákos Zahorský (from Eötvös Loránd University, Hungary)                    |
| 08/2023–05/2025               | MS student: Omar Shaikh Omar (from University of Washington, USA)                 |
| 08/2023–12/2024               | MS student: Artem Riabinin (from Lomonosov Moscow State U, Russia)                |
| 08/2023–12/2025               | PhD student: Artavazd Maranjyan (from Yerevan State U, Armenia)                   |
| 07/2023–09/2023               | Intern: Andrei Panferov (from MIPT, Russia)                                       |
| 06/2023–08/2023               | Intern: Babis Kostopoulos (from U of Athens, Greece)                              |
| 04/2023–09/2023               | Intern: Ahmad Rammal (from École Polytechnique, France)                           |
| 01/2023–05/2023               | Intern: Dinis Seward (from University of Oxford, UK)                              |
| 01/2023–12/2024               | MS student: Ivan Ilin (from Novosibirsk State University, Russia)                 |
| <b>01/2023–now</b>            | <b>PhD student: Hanmin Li (from USTC, China)</b>                                  |
| <b>09/2022–now</b>            | <b>Postdoc: Yury Demidovich (from MIPT, Russia)</b>                               |
| <b>09/2022–now</b>            | <b>PhD student: Abdurakhmon Sadiev (from MIPT, Russia)</b>                        |
| 08/2022–10/2023 <sup>25</sup> | MS student: Rafał Szlendak (from Warwick University, UK)                          |
| 08/2022–02/2023               | Intern: Wenzhi “Tom” Fang (from ShanghaiTech University, China)                   |
| 07/2022–08/2022               | Intern: Omar Shaikh Omar (from University of Washington, USA)                     |
| 07/2022–10/2022               | Intern: Michał Grudzień (from Oxford, UK)                                         |
| 06/2022–01/2023               | Intern: Artavazd Maranjyan <sup>26</sup> (from Yerevan State University, Armenia) |
| 06/2022–09/2022               | Intern: Kaja Gruntkowska (from Warwick, UK)                                       |
| <b>06/2022–now</b>            | <b>PhD student: Igor Sokolov (continuing after MS at KAUST)</b>                   |
| 01/2022–07/2022               | Intern: Abdurakhmon Sadiev (from MIPT, Russia)                                    |
| 01/2022–04/2025               | PhD student: Kai Yi (from Xi’an Jiaotong University, China)                       |
| 01/2022–06/2026               | PhD student: Grigory Malinovsky (from MIPT, Russia) <sup>27</sup>                 |
| 11/2021–06/2024               | Postdoc: Avetik Karagulyan (from CREST, France)                                   |
| 11/2021–02/2022               | Intern: Navish Kumar (from IIT Kharagpur, India)                                  |
| 09/2021–07/2026               | PhD student: Egor Shulgin (continuing after MS at KAUST)                          |
| 07/2021–11/2021               | Intern: Muhammad Harun Khan (from Imperial College, UK)                           |
| 07/2021–10/2021               | Intern: Rafał Szlendak (from Warwick University, UK)                              |
| 06/2021–06/2024               | Postdoc: Alexander Tyurin (from MIPT, Russia)                                     |
| 06/2021–08/2021               | Intern: Bokun Wang (from UC Davis, USA)                                           |
| 03/2021–06/2024               | PhD student: Lukang Sun (from Nanjing University, China)                          |
| 03/2021–08/2021               | Intern: Rustem Islamov <sup>28</sup> (from TU Munich, Germany)                    |
| 03/2021–11/2021               | Intern: Ilyas Fatkhullin <sup>29</sup> (from TU Munich, Germany)                  |
| 01/2021–11/2023               | PhD student: Slavomír Hanzely (continuing after MS at KAUST)                      |
| 09/2020–03/2022               | Research scientist: Zhize Li (from Tsinghua University, China)                    |
| 10/2020–03/2021               | Intern: Bokun Wang (from UC Davis, USA)                                           |
| 09/2020–02/2021               | Intern: Eduard Gorbunov (from MIPT, Russia)                                       |
| 08/2020–05/2025               | PhD student: Konstantin Burlachenko (from Bauman Moscow State Tech U, Russia)     |
| 08/2020–05/2022               | MS student: Igor Sokolov (from MIPT, Russia)                                      |
| 08/2020–12/2021               | MS student: Grigory Malinovsky (from MIPT, Russia)                                |
| 08/2020–09/2020               | Intern: Wenlin Chen (from University of Manchester, UK)                           |
| 06/2020–11/2020               | Intern: Rustem Islamov (from MIPT, Russia)                                        |
| 05/2020–06/2020               | Intern: Othmane Sebbouh (from École Polytechnique, France)                        |
| 05/2020–10/2020               | Intern: Ahmed Khaled Ragab (from Cairo University, Egypt)                         |
| 02/2020–12/2020               | Research scientist: El Houcine Bergou (from Toulouse, France)                     |
| 02/2020–08/2021               | MS student: Egor Shulgin (from MIPT, Russia)                                      |
| 02/2020–03/2020               | Intern: Eduard Gorbunov (from MIPT, Russia)                                       |
| 01/2020–02/2020               | Intern: Alexander Rogozin (from MIPT, Russia)                                     |

<sup>25</sup>Dropped out of MS studies to join a Large Language Model startup in Germany.

<sup>26</sup>I am supervising Arto’s MS thesis at Yerevan State University.

<sup>27</sup>next position: Postdoc, KAUST; then Postdoc, EPFL, from Nov 2026

<sup>28</sup>I have supervised Rustem Islamov’s BS thesis at MIPT. Rustem is now an MS student at Institut Polytechnique de Paris, France.

<sup>29</sup>I have supervised Ilyas Fatkhullin’s MS thesis at TU Munich. Ilyas is now a PhD student at ETH Zürich Switzerland.

|                    |                                                                                     |
|--------------------|-------------------------------------------------------------------------------------|
| 01/2020–02/2020    | Intern: Aleksandr Beznosikov (from MIPT, Russia)                                    |
| 01/2020–02/2020    | Intern: Grigory Malinovsky (from MIPT, Russia)                                      |
| 01/2020–05/2025    | PhD student: Elnur Gasanov (continuing after MS from KAUST)                         |
| 01/2020–09/2022    | PhD student: Dmitry Kovalev (continuing after MS from KAUST)                        |
| <b>11/2019–now</b> | <b>Research scientist: Laurent Condat (from Grenoble, France)</b>                   |
| 10/2019–11/2022    | Postdoc: Mher Safaryan (from Yerevan State University, Armenia)                     |
| 09/2019–08/2020    | Postdoc: Zhize Li (from Tsinghua University, China)                                 |
| 08/2019–12/2020    | MS student: Alyazeed Basyoni (from Carnegie Mellon University, USA)                 |
| 08/2019–12/2020    | MS student: Slavomír Hanzely (from Comenius University, Slovakia)                   |
| 06/2019–09/2019    | Intern: Ahmed Khaled Ragab (from Cairo University, Egypt)                           |
| 03/2019–09/2019    | Intern: Sélim Chraïbi (from Grenoble, France)                                       |
| 02/2019–10/2021    | Postdoc: Adil Salim (from Télécom ParisTech, France)                                |
| 02/2019–03/2019    | Intern: Ľudovít Horváth (from Comenius University, Slovakia)                        |
| 01/2019–02/2019    | Intern: Dmitry Kamzolov (from MIPT, Russia)                                         |
| 01/2019–02/2019    | Intern: Vladislav Elsukov (from MIPT, Russia)                                       |
| 01/2019–02/2019    | Intern: Igor Sokolov (from MIPT, Russia)                                            |
| 01/2019–02/2019    | Intern: Egor Shulgin (from MIPT, Russia)                                            |
| 01/2019–02/2019    | Intern: Eduard Gorbunov (from MIPT, Russia)                                         |
| 01/2019–03/2022    | PhD student: Alibek Sailanbayev (continuing after MS from KAUST) <sup>30</sup>      |
| 01/2019–06/2022    | PhD student: Samuel Horváth (continuing after MS from KAUST)                        |
| 11/2018–11/2021    | Postdoc: Xun Qian (from Hong Kong Baptist University, Hong Kong)                    |
| 09/2018–12/2019    | MS student: Elnur Gasanov (from MIPT, Russia)                                       |
| 09/2018–12/2019    | MS student: Dmitry Kovalev (from MIPT, Russia)                                      |
| 03/2018–08/2018    | Intern: Sarah Sachs <sup>31</sup> (from TU Munich, Germany)                         |
| 01/2018–02/2018    | Intern: Eduard Gorbunov (from MIPT, Russia)                                         |
| 01/2018–02/2018    | Intern: Elnur Gasanov (from MIPT, Russia)                                           |
| 01/2018–02/2018    | Intern: Dmitry Kovalev <sup>32</sup> (from MIPT, Russia)                            |
| 01/2018–02/2018    | Intern: Slavomír Hanzely <sup>33</sup> (from Comenius University, Slovakia)         |
| 01/2018–01/2019    | Postdoc: El Houcine Bergou (from Institut National Polytechnique, Toulouse, France) |
| 10/2017–11/2017    | Intern: Nikita Doikov (from HSE Moscow, Russia)                                     |
| 08/2017–12/2017    | PhD student: Viktor Lukáček <sup>34</sup> (from Comenius University, Slovakia)      |
| 08/2017–12/2021    | PhD student: Konstantin Mishchenko (from ENS, France)                               |
| 08/2017–11/2020    | PhD student: Filip Hanzely (now: Quant, Wincent)                                    |
| 08/2017–12/2018    | MS student: Alibek Sailanbayev (from Nazarbayev University, Kazakhstan)             |
| 08/2017–12/2018    | MS student: Samuel Horváth (from Comenius University, Slovakia)                     |
| 05/2017–05/2019    | Postdoc: Aritra Dutta (from University of Central Florida, USA)                     |
| 05/2017–07/2017    | Intern: Atal Sahu (from IIT Kanpur, India)                                          |
| 05/2017–07/2017    | Intern: Aashutosh Tiwari (from IIT Kanpur, India)                                   |

## 8.2 MY TEAM @ Kempelen Institute for Intelligent Technologies (KInIT)

|                 |                                                                |
|-----------------|----------------------------------------------------------------|
| 09/2022–10/2023 | PhD student: Ivan Agarský (from Comenius University, Slovakia) |
|-----------------|----------------------------------------------------------------|

## 8.3 MY TEAM @ Mohammed bin Zayed University of Artificial Intelligence (MBZUAI)

|                 |                                            |
|-----------------|--------------------------------------------|
| 09/2022–08/2023 | Postdoc: Sarit Khirirat (from KTH, Sweden) |
|-----------------|--------------------------------------------|

<sup>30</sup>Was forced to drop out of PhD due to serious personal/family reasons.

<sup>31</sup>I have supervised Sarah Sachs' MS thesis at TU Munich.

<sup>32</sup>I have supervised Dmitry Kovalev's BS thesis at MIPT.

<sup>33</sup>I have supervised Slavomír Hanzely's BS thesis at Comenius University.

<sup>34</sup>Viktor Lukáček left after spending 1 semester at KAUST as he realized PhD was not the right path for him.

## 8.4 MY TEAM @ MOSCOW INSTITUTE OF PHYSICS AND TECHNOLOGY

|                 |                                                                                                                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09/2020–12/2021 | PhD student: Eduard Gorbunov (PhD at MIPT, co-advised with Alexander Gasnikov; thesis: <a href="#">Distributed and Stochastic Optimization Methods with Gradient Compression and Local Steps</a> , 2021) |
| 09/2018–10/2019 | Dmitry Kamzolov                                                                                                                                                                                          |
| 09/2018–10/2019 | Vladislav Elsukov                                                                                                                                                                                        |
| 09/2018–10/2019 | Igor Sokolov (now: PhD student in my team at KAUST)                                                                                                                                                      |
| 08/2018–10/2019 | Egor Shulgin (now: postdoc in my team at KAUST)                                                                                                                                                          |
| 10/2017–10/2019 | Eduard Gorbunov (now: Postdoc at MBZUAI)                                                                                                                                                                 |
| 10/2017–08/2018 | Dmitry Kovalev (now: Postdoc at Université catholique de Louvain)                                                                                                                                        |
| 10/2017–08/2018 | Elnur Gasanov (now: PhD student in my team at KAUST)                                                                                                                                                     |

## 8.5 MY TEAM @ UNIVERSITY OF EDINBURGH

|                 |                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|
| 09/2016–07/2017 | PhD student: Filip Hanzely (transferred to KAUST after 1 year in Edinburgh to follow me, with an MS degree with distinction) |
| 03/2016–07/2016 | Postdoc: Robert M. Gower                                                                                                     |
| 10/2015–06/2019 | PhD student: Nicolas Loizou (now: Postdoc, MILA, Montréal)                                                                   |
| 10/2015–02/2017 | PhD student: Theo Pavlakou (now: Google; 2nd advisor; main advisor: Iain Murray)                                             |
| 03/2015–03/2016 | PhD student: Robert M. Gower (now: Assistant Prof. at Télécom ParisTech)                                                     |
| 03/2015–06/2015 | Visiting PhD student: Luca Bravi (from University of Florence)                                                               |
| 10/2014–03/2015 | Postdoc: Ademir Ribeiro (now: Associate Prof. at University of Paraná)                                                       |
| 09/2014–11/2017 | PhD student: Dominik Csiba (now: Algo Lead at Nozdormu, Slovakia)                                                            |
| 08/2013–07/2017 | PhD student: Jakub Konečný (now: Research Scientist, Google)                                                                 |
| 12/2013–08/2015 | Postdoc: Zheng Qu (now: Assistant Prof. at University of Hong Kong)                                                          |
| 09/2012–02/2013 | Visiting PhD student: Minnan Luo (now: Associate Prof. at Xi'an Jiaotong University)                                         |
| 10/2012–07/2014 | Postdoc: Olivier Fercoq (now: Assistant Prof. at Télécom ParisTech)                                                          |
| 02/2012–07/2014 | Postdoc: Rachael Tappenden (now: Assistant Prof. at University of Canterbury)                                                |
| 01/2012–06/2012 | Postdoc: Jakub Mareček (now: IBM Research, Dublin)                                                                           |
| 09/2010–03/2014 | PhD student: Martin Takáč (now: Associate Prof. at Mohammed bin Zayed University of Artificial Intelligence, UAE)            |
| 2010–2015       | Supervised 20 MSc Dissertations                                                                                              |
| 2010–2015       | Supervised 7 undergraduate students supported by research scholarships (EPSRC, Nuffield, College, ...)                       |

## 8.6 MY TEAM: AWARDS, RECOGNITIONS & NOTABLE ACHIEVEMENTS<sup>35</sup>

|                    |                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| 2026 (Mahran)      | Dean's List <sup>36</sup> , Applied Mathematics and Computational Science, KAUST                                       |
| 2026 (Malinovsky)  | Dean's List, Applied Mathematics and Computational Science, KAUST                                                      |
| 2026 (Shulgin)     | Dean's List, Applied Mathematics and Computational Science, KAUST                                                      |
| 2026 (Gruntkowska) | Dean's List, Statistics, KAUST                                                                                         |
| 2026 (Sadiev)      | Dean's List, Computer Science, KAUST                                                                                   |
| 2026 (Li)          | Dean's List, Computer Science, KAUST                                                                                   |
| 2026 (Sokolov)     | <b>ICLR 2026 Notable Reviewer</b>                                                                                      |
| 2026 (Malinovsky)  | Best Paper Award (runner-up), International Conference on Computational Optimization, Abu Dhabi, UAE (for paper [245]) |
| 2025 (Malinovsky)  | <b>NeurIPS 2025 Top Reviewer</b>                                                                                       |
| 2025 (Condat)      | Senior Area Editor, IEEE Transactions on Signal Processing                                                             |
| 2025 (Fradin)      | Research Internship Prize, École Polytechnique                                                                         |

<sup>35</sup>All travel grant awards are excluded.

<sup>36</sup>Carries a 2,500 USD cash prize.

|                                  |                                                                                                                           |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 2025 (Sadiev)                    | <b>2 papers accepted at NeurIPS 2025</b>                                                                                  |
| 2025 (Gruntkowska)               | <b>Google PhD Fellowship in Algorithms and Optimization<sup>37</sup> – first<sup>38</sup> in GCC</b>                      |
| 2025 (Malinovsky)                | <b>ICLR 2025 Notable Reviewer</b>                                                                                         |
| 2025 (Mahran)                    | Dean’s Award (Computer Science) <sup>39</sup> , KAUST                                                                     |
| 2025 (Malinovsky)                | Dean’s List <sup>40</sup> , Applied Mathematics and Computational Science, KAUST                                          |
| 2025 (Sokolov)                   | Dean’s List, Applied Mathematics and Computational Science, KAUST                                                         |
| 2025 (Gruntkowska)               | Dean’s List, Statistics, KAUST                                                                                            |
| 2025 (Burlachenko)               | Dean’s List, Computer Science, KAUST                                                                                      |
| 2025 (Maranjyan)                 | Dean’s List, Computer Science, KAUST                                                                                      |
| 2025 (Shulgin)                   | Dean’s List, Computer Science, KAUST                                                                                      |
| 2025 (Burlachenko)               | Accepts a Quantitative Research Scientist position at Abu Dhabi Investment Authority (ADIA), Abu Dhabi, UAE               |
| 2025 (Gasarov)                   | Accepts a Research Scientist position at Saudi Aramco, KSA                                                                |
| 2025 (Yi)                        | Accepts a Research Scientist position at META, California, USA                                                            |
| 2025 (Condat)                    | World’s Top 2% Scientist by Stanford                                                                                      |
| 2024 (Tyurin)                    | Joins Skoltech as an <b>Assistant Professor</b> & AI Research Institute (AIRI) as a <b>Team Leader</b>                    |
| 2024 (Yi)                        | Oral at NeurIPS 2024                                                                                                      |
| 2024 (Ilin)                      | Oral at NeurIPS 2024                                                                                                      |
| 2024 (Burlachenko)               | Oral at NeurIPS 2024                                                                                                      |
| 2024 (Tyurin)                    | <b>4 papers accepted at NeurIPS 2024</b>                                                                                  |
| 2024 (Tyurin)                    | Spotlight at NeurIPS 2024                                                                                                 |
| 2024 (Gruntkowska)               | Spotlight at NeurIPS 2024                                                                                                 |
| 2024 (Gruntkowska)               | Nominated by KAUST for Apple Doctoral Scholarship                                                                         |
| 2024 (Malinovsky)                | <b>ICML 2024 Best Reviewer</b>                                                                                            |
| 2024 (Gruntkowska)               | Dean’s List <sup>41</sup> , Statistics, KAUST                                                                             |
| 2024 (Malinovsky)                | Dean’s List, Applied Mathematics and Computational Science, KAUST                                                         |
| 2024 (Sadiev)                    | Dean’s List, Computer Science, KAUST                                                                                      |
| 2024 (Condat)                    | World’s Top 2% Scientist by Stanford                                                                                      |
| 2024 (Malinovsky)                | <b>NeurIPS 2024 Top Reviewer</b>                                                                                          |
| 2024 (Malinovsky)                | Summer of Reproducibility 2023 Flower AI Award <sup>42</sup> for implementing the TAMUNA algorithm (based on paper [217]) |
| 2024 (Agarský)                   | Summer of Reproducibility 2023 Flower AI Award for implementing the TAMUNA algorithm (based on paper [217])               |
| 2023 (Malinovsky)                | Nominated by KAUST for Apple PhD Scholarship <sup>43</sup>                                                                |
| 2023 (Tyurin)                    | <b>4 papers accepted at NeurIPS 2023</b>                                                                                  |
| 2023 (Sun)                       | Dean’s List <sup>44</sup> , Computer Science, KAUST                                                                       |
| 2023 (Burlachenko)               | Dean’s List, Computer Science, KAUST                                                                                      |
| 2023 (Sadiev)                    | Dean’s List, Computer Science, KAUST                                                                                      |
| 2023 (S. Hanzely)                | Dean’s List, Applied Mathematics and Computational Science, KAUST                                                         |
| 2023 (Malinovsky)                | Dean’s List, Applied Mathematics and Computational Science, KAUST                                                         |
| 2023 (Mishchenko <sup>45</sup> ) | <b>ICML 2023 Outstanding Paper Award (0.09% success rate)</b>                                                             |
| 2023 (Mishchenko)                | <b>Action Editor, Transactions on Machine Learning Research (TMLR)</b>                                                    |

<sup>37</sup>100,000 USD unrestricted gift to fund Kaja’s PhD for 2 years.

<sup>38</sup>Three more were awarded to students in the UAE in 2025.

<sup>39</sup>A financial add-on to the KAUST Fellowship, worth 6,000 USD annually, given to a few best incoming students by the Dean.

<sup>40</sup>Carries a 2,500 USD cash prize.

<sup>41</sup>In 2024, given to top 20% students at KAUST annually. Carries a 2,500 USD cash prize.

<sup>42</sup>\$ 2,000 Cash Prize shared with Ivan Agarský.

<sup>43</sup>Several invited institutions can nominate up to 3 students.

<sup>44</sup>Given to top 25% students at KAUST annually. Carries a 2,500 USD cash prize.

<sup>45</sup>Former PhD student

|                                  |                                                                                                                                                                                                                                      |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023 (Gruntkowska)               | Dean's Award (Statistics) <sup>46</sup> , KAUST                                                                                                                                                                                      |
| 2023 (Maranjyan)                 | Dean's Award (Computer Science), KAUST                                                                                                                                                                                               |
| 2023 (Mishchenko)                | Joins Samsung AI, Cambridge, UK, as a <b>Research Scientist</b>                                                                                                                                                                      |
| 2023 (Condat)                    | World's Top 2% Scientist by Stanford                                                                                                                                                                                                 |
| 2023 (Fatkhullin <sup>47</sup> ) | <b>NeurIPS 2023 Top Reviewer</b>                                                                                                                                                                                                     |
| 2022 (Kovalev)                   | <b>6 papers accepted at NeurIPS 2022</b>                                                                                                                                                                                             |
| 2022 (Burlachenko)               | Grant from AMD Inc (two GPUs)                                                                                                                                                                                                        |
| 2022 (Beznosikov <sup>48</sup> ) | <b>NeurIPS 2022 Top Reviewer<sup>49</sup></b>                                                                                                                                                                                        |
| 2022 (Tyurin)                    | <b>NeurIPS 2022 Top Reviewer</b>                                                                                                                                                                                                     |
| 2022 (Gorbunov)                  | <b>NeurIPS 2022 Top Reviewer</b>                                                                                                                                                                                                     |
| 2022 (Malinovsky)                | <b>NeurIPS 2022 Top Reviewer</b>                                                                                                                                                                                                     |
| 2022 (Sokolov)                   | <b>NeurIPS 2022 Top Reviewer</b>                                                                                                                                                                                                     |
| 2022 (Safaryan)                  | <b>NeurIPS 2022 Top Reviewer</b>                                                                                                                                                                                                     |
| 2022 (Mútny <sup>50</sup> )      | <b>NeurIPS 2022 Top Reviewer</b>                                                                                                                                                                                                     |
| 2022 (Gower <sup>51</sup> )      | <b>NeurIPS 2022 Top Reviewer</b>                                                                                                                                                                                                     |
| 2022 (Kovalev)                   | Joins Université catholique de Louvain as a <b>postdoc</b> with Yurii Nesterov                                                                                                                                                       |
| 2022 (Malinovsky)                | <b>2022 CEMSE Academic Excellence Award<sup>52</sup></b>                                                                                                                                                                             |
| 2022 (Shulgin)                   | <b>ICML 2022 Outstanding (Top 10%) Reviewer<sup>53</sup></b>                                                                                                                                                                         |
| 2022 (Gasarov)                   | <b>ICML 2022 Outstanding (Top 10%) Reviewer</b>                                                                                                                                                                                      |
| 2022 (Gorbunov)                  | <b>ICML 2022 Outstanding (Top 10%) Reviewer</b>                                                                                                                                                                                      |
| 2022 (Khaled)                    | <b>ICML 2022 Outstanding (Top 10%) Reviewer</b>                                                                                                                                                                                      |
| 2022 (Condat)                    | <b>ICML 2022 Outstanding (Top 10%) Reviewer</b>                                                                                                                                                                                      |
| 2022 (Sadiev)                    | <b>ICML 2022 Outstanding (Top 10%) Reviewer</b>                                                                                                                                                                                      |
| 2022 (Tyurin)                    | <b>ICML 2022 Outstanding (Top 10%) Reviewer</b>                                                                                                                                                                                      |
| 2022 (Sokolov)                   | <b>ICML 2022 Outstanding (Top 10%) Reviewer</b>                                                                                                                                                                                      |
| 2022 (Horváth)                   | <b>ICML 2022 Outstanding (Top 10%) Reviewer</b>                                                                                                                                                                                      |
| 2022 (Szlendak)                  | <b>ICML 2022 Outstanding (Top 10%) Reviewer</b>                                                                                                                                                                                      |
| 2022 (Mishchenko)                | <b>ICML 2022 Outstanding (Top 10%) Reviewer</b>                                                                                                                                                                                      |
| 2022 (Sadiev)                    | Dean's Award, KAUST                                                                                                                                                                                                                  |
| 2022 (Sadiev)                    | KAUST Doctoral Development Fellowship                                                                                                                                                                                                |
| 2022 (Szlendak)                  | Dean's Award (Applied Mathematics), KAUST                                                                                                                                                                                            |
| 2022 (Szlendak)                  | KAUST Doctoral Development Fellowship                                                                                                                                                                                                |
| 2022 (Shulgin)                   | Research Internship at Apple, Cambridge, UK                                                                                                                                                                                          |
| 2022 (S. Hanzely)                | Research Internship at the Flatiron Institute, New York, USA                                                                                                                                                                         |
| 2022 (S. Hanzely)                | Research Internship at the MBZUAI, Abu Dhabi, KSA                                                                                                                                                                                    |
| 2022 (Malinovsky)                | Research Internship at the CISPA Helmholtz Center for Information Security, Saarbrücken, Germany                                                                                                                                     |
| 2022 (Gorbunov)                  | Joins <b>Mohamed bin Zayed University of Artificial Intelligence</b> , Abu Dhabi, UAE, as a <b>Postdoc</b>                                                                                                                           |
| 2022 (Horváth)                   | Joins <b>Mohamed bin Zayed University of Artificial Intelligence</b> , Abu Dhabi, UAE, as an <b>Assistant Professor</b>                                                                                                              |
| 2022 (Safaryan)                  | <b>AISTATS 2022 Top Reviewer</b>                                                                                                                                                                                                     |
| 2022 (Loizou)                    | Joins <b>The Johns Hopkins University as an Assistant Professor</b> in the Department of Applied Mathematics & Statistics, and the Mathematical Institute for Data Science (MINDS), with a secondary appointment in Computer Science |

<sup>46</sup>A financial add-on to the KAUST Fellowship, worth 6,000 USD annually, given to a few best incoming students by the Dean.

<sup>47</sup>Former intern.

<sup>48</sup>Former intern.

<sup>49</sup><https://neurips.cc/Conferences/2022/ProgramCommittee>

<sup>50</sup>Former intern.

<sup>51</sup>Former PhD student.

<sup>52</sup>Carries a \$2,500 cash prize.

<sup>53</sup><https://icml.cc/Conferences/2022/Reviewers>

|                             |                                                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 (Khaled)               | Joins <b>Princeton University</b> as a PhD Student in the ECE Department <sup>54</sup>                                                                                              |
| 2022 (Gower <sup>55</sup> ) | <b>Action Editor, Transactions of Machine Learning Research (TMLR)</b>                                                                                                              |
| 2022 (Condat)               | World's Top 2% Scientist by Stanford <sup>56</sup>                                                                                                                                  |
| 2021 (Malinovsky)           | <b>2021 CEMSE Student Research Excellence Award</b> <sup>57</sup>                                                                                                                   |
| 2021 (Kovalev)              | <b>2021 CEMSE Student Research Excellence Award</b> <sup>58</sup>                                                                                                                   |
| 2021 (Horváth)              | <b>2021 Al-Kindi Statistics Research Student Award</b> <sup>59</sup>                                                                                                                |
| 2021 (Mishchenko)           | <b>Rising Stars in Data Science</b> ; invited talk to a workshop at the University of Chicago <sup>60</sup>                                                                         |
| 2021 (Gorbunov)             | <b>NeurIPS 2021 Outstanding (Top 8%) Reviewer Award</b>                                                                                                                             |
| 2021 (Mishchenko)           | <b>NeurIPS 2021 Outstanding (Top 8%) Reviewer Award</b>                                                                                                                             |
| 2021 (Shulgin)              | Research Internship at Samsung AI Research Center, Cambridge, UK                                                                                                                    |
| 2021 (Horváth)              | Research Internship at Facebook AI Research, Canada                                                                                                                                 |
| 2021 (Mishchenko)           | <b>ICML 2021 Top 10% Reviewer</b>                                                                                                                                                   |
| 2021 (Gorbunov)             | <b>ICML 2021 Top 10% Reviewer</b>                                                                                                                                                   |
| 2021 (Mishchenko)           | <b>ICML 2021 Expert Reviewer</b>                                                                                                                                                    |
| 2021 (F. Hanzely)           | <b>ICML 2021 Expert Reviewer</b>                                                                                                                                                    |
| 2021 (Gorbunov)             | <b>ICML 2021 Expert Reviewer</b>                                                                                                                                                    |
| 2021 (Condat)               | <b>ICML 2021 Expert Reviewer</b>                                                                                                                                                    |
| 2021 (Kovalev, Gasanov)     | <b>Best Student Paper Award</b> at the International Workshop on Federated Learning for User Privacy and Data Confidentiality in Conjunction with ICML 2021 (for joint paper [166]) |
| 2021 (Kovalev)              | <b>Ilya Segalovich Prize for Young Researchers</b> <sup>61</sup>                                                                                                                    |
| 2021 (Mishchenko)           | <b>Most Popular Spotlight Talk</b> (2nd place) at KAUST Conference on AI                                                                                                            |
| 2021 (Mishchenko)           | <b>ICLR 2021 Outstanding Reviewer Award</b>                                                                                                                                         |
| 2021 (Gorbunov)             | <b>ICLR 2021 Outstanding Reviewer Award</b>                                                                                                                                         |
| 2021 (Condat)               | Associate Editor, IEEE Transactions on Signal Processing                                                                                                                            |
| 2021 (Condat)               | World's Top 2% Scientist by Stanford                                                                                                                                                |
| 2020 (F. Hanzely)           | Joined <b>Toyota Tech. Institute at Chicago</b> as a Research Assistant Professor                                                                                                   |
| 2020 (Mishchenko)           | <b>2020 CEMSE Student Research Excellence Award</b> <sup>62</sup>                                                                                                                   |
| 2020 (Horváth)              | <b>Best Paper Award at the NeurIPS 2020 Workshop on Scalability, Privacy, and Security in Federated Learning</b> (for joint paper [135])                                            |
| 2020 (Loizou)               | <b>Runner Up for OR Society Best Doctoral Dissertation Prize</b> <sup>63</sup> (for year 2019)                                                                                      |
| 2020 (Horváth)              | <b>NeurIPS 2020 Best Reviewer Award</b> <sup>64</sup>                                                                                                                               |

<sup>54</sup><https://rka97.github.io>

<sup>55</sup>Former PhD student.

<sup>56</sup>[https://elsevier.digitalcommonsdata.com/datasets/btchxktzyw/4?fbclid=IwAR0u4xhKMUGIsi\\_prZLx0IOaMPzV-LNCmoILDYua90eybIVIyE6S170vyc](https://elsevier.digitalcommonsdata.com/datasets/btchxktzyw/4?fbclid=IwAR0u4xhKMUGIsi_prZLx0IOaMPzV-LNCmoILDYua90eybIVIyE6S170vyc)

<sup>57</sup>Given annually to a handful of the best Applied Mathematics students at KAUST. "The recipients exemplify the highest of academic standards and represent our confidence in your future contributions to the KAUST community, academia, and science." Carries a \$1,000 cash prize.

<sup>58</sup>Given annually to a handful of the best Computer Science students. "The recipients exemplify the highest of academic standards and represent our confidence in your future contributions to the KAUST community, academia, and science."

<sup>59</sup>Given annually to a handful of the best Statistics students at KAUST. "The recipients exemplify the highest of academic standards and represent our confidence in your future contributions to the KAUST community, academia, and science."

<sup>60</sup>The Rising Stars in Data Science workshop at the University of Chicago focuses on celebrating and fast tracking the careers of exceptional data scientists at a critical inflection point in their career: the transition to postdoctoral scholar, research scientist, industry research position, or tenure track position. An event associated with the The Center for Data and Computing (CDAC) at the University of Chicago.

<sup>61</sup>Four awards were given; each award carries a cash prize of 1,000,000 RUB ( $\approx$  14,000 USD).

<sup>62</sup>One of 5 awards given to KAUST Computer Science students.

<sup>63</sup>The OR Society was created in April 1948 as the Operational Research Club, becoming the OR Society in 1953. It is the world's oldest-established learned society catering to the OR profession and one of the largest in the world, with members in 53 countries [https://en.wikipedia.org/wiki/Operational\\_Research\\_Society](https://en.wikipedia.org/wiki/Operational_Research_Society).

<sup>64</sup><https://icml.cc/Conferences/2020/Reviewers>

|                    |                                                                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020 (Gorbunov)    | <b>NeurIPS 2020 Best Reviewer Award</b>                                                                                                                                                       |
| 2020 (F. Hanzely)  | <b>NeurIPS 2020 Best Reviewer Award</b>                                                                                                                                                       |
| 2020 (Condat)      | <b>NeurIPS 2020 Best Reviewer Award</b>                                                                                                                                                       |
| 2020 (Khaled)      | <b>NeurIPS 2020 Best Reviewer Award</b>                                                                                                                                                       |
| 2020 (Horváth)     | Research Internship at Samsung AI Research Center, Cambridge, UK                                                                                                                              |
| 2020 (Mishchenko)  | Research Internship at Google, USA (performed remotely due to Covid-19)                                                                                                                       |
| 2020 (Kovalev)     | <b>Ilya Segalovich Prize for Young Researchers</b> <sup>65</sup>                                                                                                                              |
| 2020 (Burlachenko) | Dean’s Award, KAUST                                                                                                                                                                           |
| 2020 (Malinovsky)  | Dean’s Award, KAUST                                                                                                                                                                           |
| 2020 (Mishchenko)  | <b>AAAI 2020 Outstanding Program Committee Member Award</b> (awarded to top 12 out of over 6,000 reviewers)                                                                                   |
| 2020 (Condat)      | World’s Top 2% Scientist by Stanford                                                                                                                                                          |
| 2019 (Mishchenko)  | <b>NeurIPS 2019 Best Reviewer Award</b>                                                                                                                                                       |
| 2019 (S. Hanzely)  | Dean’s Award, KAUST                                                                                                                                                                           |
| 2019 (F. Hanzely)  | Research Internship at Google, New York                                                                                                                                                       |
| 2019 (Horváth)     | Research Internship at Amazon, Berlin                                                                                                                                                         |
| 2019 (Sailanbayev) | Research Internship at Intel, Germany                                                                                                                                                         |
| 2018 (Kovalev)     | Dean’s Award, KAUST                                                                                                                                                                           |
| 2018 (Loizou)      | Research Internship at Facebook AI Research (FAIR), Montréal                                                                                                                                  |
| 2018 (Mishchenko)  | Research Internship at Amazon, Seattle                                                                                                                                                        |
| 2018 (F. Hanzely)  | Research Internship at Microsoft Research (with Lin Xiao)                                                                                                                                     |
| 2018 (F. Hanzely)  | Research Internship at Amazon, Berlin, Scalable Machine Learning Group                                                                                                                        |
| 2018 (Horváth)     | <b>Best DS<sup>3</sup> Poster Award</b> <sup>66</sup> , Paris (1st Prize; for joint paper [81])                                                                                               |
| 2018 (Doikov)      | <b>Best Talk Award</b> <sup>67</sup> , Voronovo, Russia (1st Prize; for joint paper [69])                                                                                                     |
| 2018 (F. Hanzely)  | WEP Best Poster Award (3rd Place), KAUST                                                                                                                                                      |
| 2017 (Mishchenko)  | Dean’s Award, KAUST                                                                                                                                                                           |
| 2017 (Lukáček)     | Dean’s Award, KAUST                                                                                                                                                                           |
| 2017 (F. Hanzely)  | Dean’s Award, KAUST                                                                                                                                                                           |
| 2017 (Gower)       | <b>18th IMA Leslie Fox Prize</b> <sup>68</sup> (2nd Prize; for joint paper [39])                                                                                                              |
| 2016 (Csiba)       | Postgraduate Essay Prize, School of Mathematics, University of Edinburgh                                                                                                                      |
| 2016 (F. Hanzely)  | <b>CASE PhD Studentship</b> (£93,333 award; 3/4 from EPSRC, 1/4 from Amazon)                                                                                                                  |
| 2016 (Loizou)      | A. G. Leventis Foundation Grant for PhD studies                                                                                                                                               |
| 2015 (Takáč)       | <b>OR Society Best Doctoral Dissertation Prize</b> (for year 2014)                                                                                                                            |
| 2015 (Loizou)      | A. G. Leventis Foundation Grant for PhD studies                                                                                                                                               |
| 2015 (Loizou)      | Principal’s Career Development Scholarship <sup>69</sup> (in Data Science)                                                                                                                    |
| 2015 (Kisiala)     | Best Student Prize <sup>70</sup> , OR MSc Programme, School of Mathematics, Edinburgh                                                                                                         |
| 2015 (Fercoq)      | <b>17th IMA Leslie Fox Prize</b> (2nd Prize; for joint paper [21])                                                                                                                            |
| 2015 (Csiba)       | <b>Best Contribution Award</b> (2nd Prize; for joint paper [35]), Workshop: Optimization and Big Data, Edinburgh. Committee: Arkadi Nemirovskii (Georgia Tech) and Rodolphe Jenatton (Amazon) |
| 2015 (Konečný)     | <b>BASP Frontiers Best Contribution Award</b> (1st prize in the field of signal processing; for joint paper [20]), Villars-sur-Ollon, Switzerland                                             |

<sup>65</sup>Nine awards were given in the area of Computer Science; each award carries a cash prize of 350,000 RUB ( $\approx$  5,000 USD).

<sup>66</sup>DS<sup>3</sup> stands for Data Science Summer School, held at École Polytechnique, Paris, during June 25–29, 2018. There were 170 posters in the competition, from MS and PhD students, and postdocs. Samuel’s poster, based on joint work [81], won the main prize, which also attracted a 500 EUR check.

<sup>67</sup>Event: “Traditional Youth School in Control, Information and Optimization”, organized by Boris Polyak.

<sup>68</sup>“The Leslie Fox Prize is a biennial prize established in 1985 by the IMA in honour of mathematician Leslie Fox (1918-1992). The prize honours young numerical analysts worldwide (any person less than 31 years old), and applicants submit papers for review. A committee [...] awards First Prize and Second Prizes based on mathematical and algorithmic brilliance in tandem with presentational skill”

<sup>69</sup>Principal’s Career Development Scholarship: A highly competitive scholarship offered to 3 incoming PhD students in mathematics at the University of Edinburgh each year.

<sup>70</sup>For best performance in courses and MSc Dissertation, which I supervised.

|                     |                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2014 (Konečný)      | <b>Google European Doctoral Fellowship</b> <sup>71</sup> (\$180,000 unrestricted gift funding Jakub's PhD for 3 years)                                               |
| 2014 (Csiba)        | Principal's Career Development Scholarship (in Data Science)                                                                                                         |
| 2013 (Konečný)      | Principal's Career Development Scholarship (in Data Science)                                                                                                         |
| 2013 (Takáč)        | <b>16th IMA Leslie Fox Prize</b> (2nd Prize; for joint paper [10])                                                                                                   |
| 2013 (Takáč)        | SIAM Certificate in Recognition of Outstanding Efforts and Accomplishments, on behalf of the SIAM Chapter at the University of Edinburgh for academic year 2012–2013 |
| 2012 (Takáč)        | <b>INFORMS Computing Society Best Student Paper Prize</b> (sole runner up; for joint paper [8]), Phoenix, Arizona                                                    |
| 2012 (Banks-Watson) | Best Student Prize, OR MSc Programme, School of Mathematics, Edinburgh                                                                                               |
| 2012 (Takáč)        | <b>Best Talk Award</b> , SIAM National Student Chapter Conference, Manchester, UK                                                                                    |
| 2012 (Takáč)        | Best Talk Award, Edinburgh Postgraduate Colloquium, University of Edinburgh                                                                                          |
| 2012 (Takáč)        | Alice Margaret Campbell Bequest Fund Award for success in 1st year of PhD                                                                                            |
| 2011 (Takáč)        | Certificate of Appreciation, 24th Biennial Conf. on Numerical Analysis, Glasgow, UK                                                                                  |
| 2011 (Takáč)        | Best Poster Award, SIAM Student Chapter Conference, Edinburgh, UK                                                                                                    |

## 8.7 MY TEAM: SELECTED PRIOR/INDEPENDENT ACHIEVEMENTS<sup>72</sup>

|                    |                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------|
| 2026 (Loizou)      | <b>NSF CAREER Award</b> <sup>73</sup>                                                            |
| 2026 (Gower)       | Outstanding Paper Honorable Mention Award <sup>74</sup> at ICLR 2026                             |
| 2025 (Gorbunov)    | Area Chair, NeurIPS                                                                              |
| 2024 (Gorbunov)    | Action Editor, TMLR                                                                              |
| 2024 (Mishchenko)  | Action Editor, TMLR                                                                              |
| 2024 (Takáč)       | Senior Area Chair, NeurIPS                                                                       |
| 2023 (Gower)       | Action Editor, TMLR                                                                              |
| 2023 (Záhorský)    | Member, Slovak Committee of Mathematical Olympiad                                                |
| 2022 (Záhorský)    | Coordinator, European Girls' Mathematical Olympiad, Hungary                                      |
| 2022 (Condat)      | Meritorious Service Award from the journal Mathematical Programming <sup>75</sup>                |
| 2022 (Burlachenko) | Second Place, KAUST Chess Tournament                                                             |
| 2021 (Gaponov)     | Bronze Medal, Belarusian National Olympiad in Competitive Programming                            |
| 2021 (Maranjyan)   | Outstanding Final Project Award <sup>76</sup> , Yerevan State University, Armenia                |
| 2020 (Záhorský)    | Deputy Leader, 14th Middle European Mathematical Olympiad, virtual                               |
| 2020 (Basyoni)     | National Deputy Leader and Head Coach at the International Olympiad of Informatics, Saudi Arabia |
| 2019 (Záhorský)    | Deputy Leader, 13th Middle European Mathematical Olympiad, Pardubice, Czech Republic             |
| 2019 (Panferov)    | Gold Medal, International Physics Olympiad, Tel Aviv, Israel                                     |
| 2019 (Riabinin)    | Winner, Phystech Olympiad in Physics, Dolgoprudny, Russia                                        |

<sup>71</sup>Google quote: "Nurturing and maintaining strong relations with the academic community is a top priority at Google. Today, we're announcing the 2014 Google PhD Fellowship recipients. These students, recognized for their incredible creativity, knowledge and skills, represent some of the most outstanding graduate researchers in computer science across the globe. We're excited to support them, and we extend our warmest congratulations."

<sup>72</sup>These awards are independent of my input, and were in most cases obtained before joining my team.

<sup>73</sup>Project: "Structural and Algorithmic Foundations of Variational Inequalities in Machine Learning: From Principles to Scalability". [https://www.nsf.gov/awardsearch/show-award/?AWD\\_ID=2542902](https://www.nsf.gov/awardsearch/show-award/?AWD_ID=2542902)

<sup>74</sup>Two papers won the Main Prize, and one paper won the Honorable Mention Award. More than 19,000 papers were submitted to ICLR in 2026.

<sup>75</sup>Mathematical Programming is the leading optimization journal. "The Meritorious Service Award was created to acknowledge these continued efforts. In 2022 our Editorial Board assessed the referees who have demonstrated exceptional diligence in their service to the journal."

<sup>76</sup>Awarded to 6 students from more than 250 students for best undergraduate thesis.

|                                 |                                                                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 2019 (Basyoni)                  | National Deputy Leader and Head Coach at the International Olympiad of Informatics, Saudi Arabia                   |
| 2019 (Li)                       | Tsinghua Outstanding Doctoral Dissertation Award                                                                   |
| 2017 (Záhorský)                 | Silver Medal, 59th International Mathematical Olympiad, Cluj-Napoca, Romania (representing Slovakia)               |
| 2018 (Riabinin)                 | Prizewinner, Regional Stage of the All Russian Olympiad in Physics, Russia                                         |
| 2018 (Riabinin)                 | 2nd Place, City Physics Olympiad, Nizhny Novgorod, Russia                                                          |
| 2018 (Riabinin)                 | Prizewinner, Phystech Olympiad in Mathematics, Dolgoprudny, Russia                                                 |
| 2018 (Mishchenko & Sailanbayev) | 80th Place, 2018 IEEEExtreme programming competition <sup>77</sup>                                                 |
| 2017 (Záhorský)                 | Honorable Mention, 58th International Mathematical Olympiad, Rio de Janeiro, Brazil (representing Slovakia)        |
| 2017 (Ilin)                     | Captain of the Russian team @ International Young Physicists Tournament, Singapore/Russia                          |
| 2017 (Karagulyan)               | Second Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria                |
| 2017 (S. Hanzely)               | 8–10th Place, Vojtech Jarník International Mathematical Competition (1st place among Czech and Slovak contestants) |
| 2017 (Horváth)                  | 37th Place, Vojtech Jarník International Mathematical Competition, Ostrava, Czech Republic                         |
| 2016 (Záhorský)                 | Silver Medal, 10th Middle European Mathematical Olympiad, Vöcklabruck, Austria (representing Slovakia)             |
| 2016 (Karagulyan)               | 2nd Prize, Mirror of William Lowell Putnam Mathematical Competition                                                |
| 2016 (Malinovsky)               | Abramov's Scholarship for students with the best grades at MIPT                                                    |
| 2016 (S. Hanzely)               | Participation, 57th International Mathematical Olympiad, Hong Kong                                                 |
| 2016 (S. Hanzely)               | 3rd Place, Slovak National Mathematical Olympiad                                                                   |
| 2016 (S. Hanzely)               | 1st Place, Slovak Mathematical Olympiad, Regional Round                                                            |
| 2016 (S. Hanzely)               | 1st Place, Slovak Informatics Olympiad, Regional Round                                                             |
| 2016 (Horváth)                  | 36th Place, Vojtech Jarník International Mathematical Competition, Ostrava, Czech Republic                         |
| 2015 (Karagulyan)               | Third Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria                 |
| 2016 (Horváth)                  | 3rd Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria                   |
| 2016 (Sailanbayev)              | Semifinalist, ACM ICPC Programming Contest, NEERC region, Almaty, Kazakhstan                                       |
| 2015 (Karagulyan)               | Second Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria                |
| 2015 (Karagulyan)               | Semifinalist, ACM-ICPC Programming Contest, NEERC region, Tbilisi, Georgia                                         |
| 2015 (S. Hanzely)               | Bronze Medal, Middle European Mathematical Olympiad                                                                |
| 2015 (S. Hanzely)               | 2nd Place, Slovak Informatics Olympiad, Regional Round                                                             |
| 2015 (Sailanbayev)              | 2nd Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria                   |
| 2015 (Mishchenko)               | 1st Prize, HSE Olympiad in Applied Mathematics and Informatics, Moscow, Russia                                     |
| 2014 (Karagulyan)               | Semifinalist, ACM-ICPC Programming Contest, NEERC region, Tbilisi, Georgia                                         |
| 2014 (Malinovsky)               | Bronze Medal, International Zhautykov Olympiad in Physics                                                          |
| 2014 (Malinovsky)               | Participant, All-Russian Physics Olympiad                                                                          |
| 2014 (S. Hanzely)               | 1st Place, Slovak Mathematical Olympiad, Regional Round                                                            |
| 2014 (Kovalev)                  | Honorable Mention, 15th Asian Physics Olympiad, Singapore                                                          |
| 2014 (Kovalev)                  | Winner, All Russian Mathematics Olympiad (Moscow Region)                                                           |
| 2014 (Kovalev)                  | Winner, All Russian Computer Science Olympiad (Moscow Region)                                                      |
| 2014 (Kovalev)                  | Prizewinner, All Russian Physics Olympiad                                                                          |

<sup>77</sup>4,000 teams (of size 3) from all over the world competed in a 24-hour time span against each other to solve a set of programming problems. Konstantin and Alibek scored high despite being just 2 on the team!

|                    |                                                                                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2014 (Mishchenko)  | 3rd Prize, MIPT Student Mathematical Olympiad, Moscow, Russia                                                                                                                |
| 2014 (Horváth)     | 18th Place, National Mathematical Olympiad, Bratislava, Slovakia                                                                                                             |
| 2014 (Horváth)     | 1st Place, Nitra Region Mathematical Olympiad, Category A, Slovakia                                                                                                          |
| 2014 (Sailanbayev) | 2nd Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria                                                                             |
| 2014 (Loizou)      | Top 1% in Mathematics at National and Kapodestrian University of Athens, Greece                                                                                              |
| 2014 (Csiba)       | Best Student Work in Applied Informatics in Czech and Slovak Republic, Annual Student Scientific Conference, Ústí nad Labem, Czech Republic                                  |
| 2014 (F. Hanzely)  | 2nd Prize (101st place), International Mathematical Competition for University Students, Blagoevgrad, Bulgaria                                                               |
| 2014 (F. Hanzely)  | 9th Place, V. Jarník International Mathematical Competition, Ostrava, Czech Republic                                                                                         |
| 2014 (Lukáček)     | 26th Place, Vojtech Jarník International Mathematical Competition, Ostrava, Czech Republic                                                                                   |
| 2013 (Karagulyan)  | Semifinalist, ACM-ICPC Programming Contest, NEERC region, Tbilisi, Georgia                                                                                                   |
| 2013 (Karagulyan)  | 2nd Prize, Mirror of William Lowell Putnam Mathematical Competition                                                                                                          |
| 2013 (Malinovsky)  | Prizewinner, All-Russian Physics Olympiad                                                                                                                                    |
| 2013 (S. Hanzely)  | 1st Place, Slovak Mathematical Olympiad, Regional Round                                                                                                                      |
| 2013 (Kovalev)     | Winner, All Russian Physics Olympiad                                                                                                                                         |
| 2013 (Sailanbayev) | Silver Medal, International Mathematical Olympiad, Santa Marta, Colombia                                                                                                     |
| 2013 (F. Hanzely)  | Bronze Medal, International Mathematical Olympiad, Santa Marta, Colombia                                                                                                     |
| 2013 (Karagulyan)  | Honourable Mention, International Mathematical Olympiad, Santa Marta, Colombia                                                                                               |
| 2013 (Sailanbayev) | 1st Place, National Mathematical Olympiad, Kazakhstan                                                                                                                        |
| 2013 (F. Hanzely)  | 1st Place, Slovak National Round of Mathematical Olympiad, Košice, Slovakia                                                                                                  |
| 2013 (Sailanbayev) | Gold Medal, International Zhaautykov Olympiad, Almaty, Kazakhstan                                                                                                            |
| 2013 (Lukáček)     | 20th Place, Vojtech Jarník International Mathematical Competition, Ostrava, Czech Republic                                                                                   |
| 2012 (Karagulyan)  | Honourable Mention, International Mathematical Olympiad, Mar del Plata, Argentina                                                                                            |
| 2012 (Kovalev)     | Prizewinner, All Russian Physics Olympiad                                                                                                                                    |
| 2012 (Lukáček)     | 3rd Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria                                                                             |
| 2012 (Mishchenko)  | 1st Prize, Moscow Mathematical Olympiad, Moscow, Russia                                                                                                                      |
| 2012 (Mishchenko)  | 1st Prize, PhysTech International Olympiad in Mathematics                                                                                                                    |
| 2012 (Basyoni)     | Silver Medal <sup>78</sup> , International Mathematical Olympiad, Mar del Plata, Argentina                                                                                   |
| 2012 (Sailanbayev) | Bronze Medal, International Mathematical Olympiad, Mar del Plata, Argentina                                                                                                  |
| 2012 (Sailanbayev) | Silver Medal, Balkan Mathematical Olympiad, Antalya, Turkey                                                                                                                  |
| 2012 (F. Hanzely)  | Bronze Medal, Middle European Mathematical Olympiad, Solothurn, Switzerland                                                                                                  |
| 2012 (Csiba)       | FIDE International Master in Chess                                                                                                                                           |
| 2012 (Csiba)       | 3rd Prize, International Mathematical Competition, Blagoevgrad, Bulgaria                                                                                                     |
| 2012 (Konečný)     | 2nd Prize, International ChaLearn Competition, One shot learning of gestures from Microsoft Kinect videos                                                                    |
| 2012 (Fercoq)      | Gaspard Monge Prize “for best PhD thesis defended in France 2012 in mathematics or computer science, with significant contributions to Optimization and Operations Research” |
| 2012 (Luo)         | Google Anita Borg Scholarship, China                                                                                                                                         |
| 2012 (Lukáček)     | 2nd Place, International Correspondence Seminar in Mathematics (iKS)                                                                                                         |
| 2011 (Lukáček)     | Bronze Medal (26th Place), Middle European Mathematical Olympiad, Varaždin, Croatia                                                                                          |
| 2010 (Konečný)     | Honourable Mention, International Mathematical Olympiad, Astana, Kazakhstan                                                                                                  |
| 2010 (Csiba)       | Honourable Mention, Middle European Mathematical Olympiad, Žilina, Slovakia                                                                                                  |
| 2008 (Konečný)     | Honourable Mention, Middle European Mathematical Olympiad, Olomouc, Czech Republic                                                                                           |

<sup>78</sup>Historically the first silver medal at IMO by Saudi Arabia.

2007–2009 (Takáč)      Winner, 3rd Place and Honourable Mention (twice), International Student Scientific Conference, Czech and Slovak Republic

## 9. TALKS

### 9.1 TALKS: SUMMARY

I have given **more than 270 research talks**<sup>79</sup> at conferences, workshops and seminars worldwide (Australia, Austria, Belgium, Brazil, Canada, Chile, China, Cuba, France, Germany, Greece, Hong Kong, Hungary, India, Japan, Mongolia, Morocco, Netherlands, Portugal, Russia, Saudi Arabia, Slovakia, Slovenia, Spain, Switzerland, UAE, United Kingdom, Uruguay, USA). Out of these, **≈80 are plenary talks** at conferences and workshops, **≈30 are invited PhD courses and tutorials**, **≈100 are seminar talks**, and the rest are invited and contributed conference talks. I regularly give talks at the premier international optimization conferences (each taking place once in 3 years): Int. Symposium on Mathematical Programming (Rio'06, Chicago'09, Berlin'12, Pittsburgh'15, Bordeaux'18), SIAM Conf. on Optimization (Darmstadt'11, San Diego'14, Vancouver'17, Hong Kong'20), Int. Conf. on Continuous Optimization (Ontario'07, Santiago'10, Lisbon'13, Tokyo'16, Berlin'19).

### 9.2 PLENARY TALKS<sup>80</sup>

|         |                                                                                                                                                                                                   |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12/2026 | <b>CODEC-FM: Collaborative, Open, and DECentralized training of Foundation Models</b> , NeurIPS 2026 Workshop, Sydney, Australia                                                                  |
| 11/2026 | <b>International Conference on Computational Optimization (ICOMP 2026)</b> , Casablanca, Morocco                                                                                                  |
| 09/2026 | <b>1st Conference and Workshop on Mathematics and Artificial Intelligence</b> , Nazarbayev University, Astana, Kazakhstan                                                                         |
| 06/2026 | <b>The 9th International Conference on Dynamics of Information Systems (DIS 2026)</b> , Ústí nad Labem, Czech Republic                                                                            |
| 05/2026 | <b>ELLIIT Symposium on Optimization for Learning</b> , Lund, Sweden                                                                                                                               |
| 02/2026 | <b>Workshop on Learning from Heterogeneous Sources</b> , Simons Institute, Berkeley, USA                                                                                                          |
| 11/2025 | <b>KAUST Workshop on Distributed Training in the Era of Large Models</b> , KAUST, Thuwal, Saudi Arabia                                                                                            |
| 10/2025 | <b>The 3rd IEEE International Conference on Federated Learning Technologies and Applications (FLTA 2025)</b> , Dubrovnik, Croatia                                                                 |
| 09/2025 | <b>10th Workshop on High-Dimensional Approximation (HDA2025)</b> , Bonn, Germany                                                                                                                  |
| 08/2025 | <b>International Workshop on Secure and Efficient Federated Learning In Conjunction with ACM AsiaCCS 2025 (FL-AsiaCCS'25)</b> , Hanoi, Vietnam                                                    |
| 06/2025 | <b>3rd Vienna Workshop on Computational Optimization 2025 (VWCO25)</b> , Vienna, Austria                                                                                                          |
| 04/2025 | <b>Rising Stars in AI Symposium</b> , Thuwal, KAUST, Saudi Arabia                                                                                                                                 |
| 10/2024 | <b>International Conference on Computational Optimization (ICOMP-2024)</b> , Innopolis, Russian Federation                                                                                        |
| 09/2024 | <b>2nd IEEE International Conference on Federated Learning Technologies and Applications (FLTA 2024)</b> , Valencia, Spain (virtual)                                                              |
| 08/2024 | <b>ALGOPT2024 workshop on Algorithmic Optimization: Tools for AI and Data Science</b> , Louvain-la-Neuve, Belgium. Celebration of Yurii Nesterov's 50 years long research career in optimization. |

<sup>79</sup>All my talks are listed on [https://richtarik.org/i\\_talks.html](https://richtarik.org/i_talks.html)

<sup>80</sup>For the purpose of this CV, a plenary talk is any talk not given to a sub-audience; or a talk explicitly labeled as a plenary/keynote talk by the organizers of the workshop/conference. I am excluding here talks at events I organized or co-organized and declined invites to deliver a plenary talk. I am including past talks, and accepted invites to give a talk.

06/2024 **Federated Learning for Computer Vision (FedVision) Workshop**, CVPR, Seattle, USA (virtual)

06/2024 **Applied Algorithms for Machine Learning – a Workshop on the Future of Computation**, Paris, France

06/2024 **One World Optimization Seminar in Vienna**, Erwin Schrödinger International Institute for Mathematics and Physics, Vienna, Austria

04/2024 **Workshop on Nonsmooth Optimization and Applications (NOPTA 2024)**, In Honor of the 75th Birthday of Boris Mordukhovich, University of Antwerp, Belgium

02/2024 **Apple Workshop on Privacy Preserving Machine Learning**, Cupertino, California, USA (virtual)

12/2023 **NeurIPS 2023 Workshop on Federated Learning in the Age of Foundation Models**, New Orleans, Louisiana, USA

07/2023 **ICML 2023 Workshop. Federated Learning and Analytics in Practice: Algorithms, Systems, Applications, and Opportunities**, Honolulu, Hawaii

07/2023 **Federated and Collaborative Learning Workshop**, Simons Institute, Berkeley, USA

07/2023 **Mathematics in Armenia: Advances and Perspectives** (80th anniversary of the foundation of the Armenian National Academy of Sciences), Yerevan, Armenia

12/2022 **Optimization in the Big Data Era**, Institute of Mathematical Sciences, National University of Singapore, Singapore, Optimization in the Big Data Era, Institute for Mathematical Sciences, National University of Singapore, Singapore

11/2022 **KAUST Workshop on Scientific Computing and Machine Learning**, KAUST

11/2022 **Google’s 2022 Workshop on Federated Learning and Analytics**, virtual

10/2022 **MBZUAI Workshop on Collaborative Learning: From Theory to Practice**, Abu Dhabi (invited by Michael I. Jordan)

09/2022 **CrossFL: Cross-Community Federated Learning: Algorithms, Systems and Co-designs**, workshop associated with the MLSys conference, Santa Clara, USA (virtual)

06/2022 **Mathematics of Complex Data**, KTH Royal Institute of Technology, Stockholm, Sweden

05/2022 **Workshop on Stochastic Numerics, Statistical Learning, Optimization, Approximations, with Applications**, KAUST, Saudi Arabia

04/2022 **Lagrange Workshop on Federated Learning**, Lagrange Mathematics and Computing Research Center, virtual

04/2022 **Apple’s Workshop on Privacy Preserving Machine Learning**, virtual

02/2022 **Dagstuhl Seminar**, Theory of Randomized Optimization Heuristics, 3 talks, Germany

12/2021 **NeurIPS 2021 Workshop. New Frontiers in Federated Learning: Privacy, Fairness, Robustness, Personalization and Data Ownership**, Virtual

11/2021 **KAUST-GSAI Joint Workshop on Advances in AI**, Virtual

11/2021 **Google Federated Learning and Analytics Workshop**, Virtual

07/2021 **Optimization Without Borders** (celebration of the 65th Birthday of Yurii Nesterov), Sirius University, Sochi, Russia (virtual)

04/2021 **KAUST Conference on Artificial Intelligence**, 2 keynote talks, Thuwal, Saudi Arabia

08/2020 **Workshop on Privacy Preserving Machine Learning**, Apple, Virtual Workshop

07/2020 **ICML 2020 Workshop: Beyond First Order Methods in ML Systems**, Virtual

06/2020 **Mathematics of Data Science**, Virtual Conference, United Kingdom

10/2019 **School-Conference “Approximation and Data Analysis”**, Nizhny Novgorod, Russia

09/2019 **Workshop on Optimization, Statistics and Numerical Methods**, Moscow Institute of Physics and Technology, Dolgoprudny, Russia (workshop organized around my visit to MIPT)

09/2019 **50 Years of Mathematics in Bielefeld - the (new) Unity of Mathematics**, Bielefeld, Germany

09/2019 **DIMACS Workshop on Randomized Numerical Linear Algebra, Statistics, and Optimization**, Rutgers University, USA

06/2019 **Approximation, Sampling, and Compression in High Dimensional Problems**, Isaac Newton Institute for Mathematical Sciences Program on “Approximation, Sampling and Compression in Data Science”, Cambridge University, UK

- 02/2019 **Numerical Algorithms in Nonsmooth Optimization**, Thematic Program: “Modern Maximal Monotone Operator Theory: From Nonsmooth Optimization to Differential Inclusions”, Erwin Schrödinger International Institute for Mathematics and Physics (ESI), Vienna, Austria
- 11/2018 **Statistics and Data Science Workshop**, KAUST, Thuwal, KSA
- 09/2018 **Randomized Numerical Linear Algebra and Applications**, Program: Data Science, Simons Institute, Berkeley, USA
- 08/2018 **DIMACS/TRIPODS Workshop: Optimization in Machine Learning**, Lehigh University, Bethlehem, USA
- 07/2018 **XII Brazilian Workshop on Continuous Optimization**, Foz do Iguaçu, Brazil
- 10/2017 **Optimization at Work**<sup>81</sup>, Moscow Institute of Physics and Technology, Moscow, Russia
- 09/2017 **Workshop on Decentralized Machine Learning, Optimization and Privacy**, Lille, France
- 07/2017 **Workshop on Convex Optimization and Applications**, Fields Institute, Toronto, Canada (in honour of 70th birthday of Arkadi Nemirovski)
- 04/2017 **Visual Computing - Modeling and Reconstruction**, KAUST, Thuwal, KSA
- 01/2017 **2017 BASP Frontiers Workshop**, Villars-sur-Ollon, Switzerland
- 11/2016 **Workshop on Distributed Machine Learning**, Telecom ParisTech, Paris, France
- 11/2016 **SIAM Warwick Student Chapter Conference on Machine Learning and Statistics**, Coventry, UK
- 10/2016 **41st Woudschoten Conference**, Zeist, Netherlands. Two keynote lectures in the stream “Numerical methods for big data analytics”
- 09/2016 **Linear Algebra and Parallel Computing at the Heart of Scientific Computing**, a joint event of the Royal Society of Edinburgh and the French Embassy in London, Edinburgh, UK
- 09/2016 **“OR58”: The 58th Annual Conference of the Operational Research Society**, Portsmouth, UK (closing plenary)
- 06/2016 **2016 Int. Workshop on Modern Optimization and Applications (MOA 2016)**, Beijing, China
- 04/2016 **Einstein Center Mathematical Colloquium “Sparsity: Statistics, Optimization, and Applications”**, Berlin, Germany. “The purpose of this biannual scientific colloquium is bringing together mathematicians, scientists, and engineers to enjoy a series of talks on one topical issue of current or emerging interest to several fields within mathematics.”
- 03/2016 **Computationally and Statistically Efficient Inference for Complex Large-scale Data**, Oberwolfach, Germany
- 09/2015 **Statistical and Computational Challenges in Large-Scale Data Analysis**, Alan Turing Institute Workshop, Cambridge, UK
- 09/2015 **LMS Inverse Day: Large-Scale and Nonlinear Inverse Problems**, Edinburgh, UK
- 04/2015 **Maxwell Institute Probability Day**, Edinburgh, UK
- 01/2015 **Optimization and Statistical Learning**, Les Houches, France
- 01/2015 **Theory of Big Data Science**, University College London, UK
- 12/2014 **Optimization Workshop, Foundations of Computational Mathematics**, Montevideo, Uruguay
- 11/2014 **46th Conference of Slovak Mathematicians**, Jasná, Slovakia
- 09/2014 **Mathematical Methods in Economics and Engineering**, Smolenice, Slovakia
- 07/2014 **Understanding Complex and Large Industrial Data**, Lancaster, UK
- 05/2014 **9th Int. Conf. on Intelligent Systems: Theories and Applications**, Rabat, Morocco
- 02/2014 **Stochastic Gradient Methods**, Inst. for Pure and Applied Mathematics, Los Angeles, USA
- 12/2013 **NeurIPS Workshop on Optimization in Machine Learning**, Lake Tahoe, USA. Past plenary speakers: D. Bertsekas, L. Bottou, S. Wright (2008), N. Srebro, L. Vandenberghe, A. Nemirovski (2009), M. Schmidt, Yu. Nesterov (2010), B. Recht, S. Boyd (2011), P. Parillo, F. Bach (2012)

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<sup>81</sup>This event was organized in my honour.

|         |                                                                                                                                                                   |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11/2013 | <b>International Conference on Information Technologies and Society</b> , Slovenia                                                                                |
| 10/2013 | <b>Parallel and Distributed Algorithms for Inference and Optimization</b> , Simons Institute for the Theory of Computing, University of California, Berkeley, USA |
| 05/2013 | <b>Big Data Mining</b> , Imperial College London, UK                                                                                                              |
| 03/2013 | <b>Fête Parisienne in Computation, Inference and Optimization</b> , IHES, Paris, France                                                                           |
| 03/2013 | <b>Edinburgh SIAM Student Chapter Conference</b> , Edinburgh, UK                                                                                                  |
| 02/2013 | <b>Big Data and Social Media</b> , Glasgow, UK                                                                                                                    |
| 01/2013 | <b>Optimization and Statistical Learning</b> , Les Houches, France                                                                                                |
| 07/2012 | <b>Optimization in Machine Learning</b> , ICML workshop, Edinburgh, UK                                                                                            |
| 07/2011 | <b>Optimization Workshop, Foundations of Comp. Mathematics</b> , Budapest, Hungary                                                                                |
| 05/2011 | <b>Computational Complexity Challenges in Optimization</b> , Edinburgh, UK                                                                                        |

### 9.3 INVITED LECTURE SERIES, TUTORIALS & SUMMER SCHOOL COURSES

|         |                                                                                                                                                                                    |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01/2026 | <b>Federated Optimization Tutorial</b> (3.5 hours), Bootcamp, Simons Institute Program on Federated and Collaborative Learning, Berkeley, USA                                      |
| 11/2025 | <b>Introduction to Machine Learning 2</b> (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia                                                                               |
| 10/2025 | <b>Introduction to Machine Learning 1</b> (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia                                                                               |
| 07/2025 | <b>The Machine Learning Summer School (MLSS 2025)</b> , MBour, Senegal                                                                                                             |
| 07/2025 | <b>Introduction to Optimization 2</b> (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia                                                                                   |
| 06/2025 | <b>Introduction to Optimization 1</b> (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia                                                                                   |
| 12/2024 | <b>Optimization for Machine Learning</b> (three lectures), Applied Mathematics School, KAUST, Saudi Arabia                                                                         |
| 11/2024 | <b>Optimization for Machine Learning</b> (five 90 min lectures), Beijing Institute for Mathematical Sciences and Applications (BIMSA), Beijing, China                              |
| 02/2024 | <b>Optimization for Machine Learning</b> (three 50 min lectures), AMCS-STAT School, KAUST, Saudi Arabia                                                                            |
| 03/2024 | Machine Learning Summer School, Okinawa, Japan (declined due to a clash with annual leave)                                                                                         |
| 07/2023 | Eastern European Machine Learning Summer School (EEML 2023), Košice, Slovakia                                                                                                      |
| 06/2023 | <b>Introduction to Machine Learning 2</b> (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia                                                                               |
| 06/2023 | <b>Introduction to Machine Learning 1</b> (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia                                                                               |
| 07/2023 | <b>Eastern European Machine Learning Summer School</b> , Košice, Slovakia                                                                                                          |
| 11/2022 | <b>Introduction to Optimization 2</b> (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia                                                                                   |
| 11/2022 | <b>Introduction to Optimization 1</b> (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia                                                                                   |
| 06/2022 | <b>Introduction to Stochastic Gradient Descent Methods</b> (PhD course, 22.5 hours), School of Mathematics, Physics and Informatics, Bratislava, Slovakia                          |
| 06/2022 | <b>Introduction to Stochastic Gradient Descent Methods</b> (PhD course, 18 hours), Vienna Graduate School for Computational Optimization (VGSCO), Vienna, Austria                  |
| 10/2019 | <b>A Guided Walk Through the ZOO of Stochastic Gradient Descent Methods</b> (Mini-course, 2.5 hours), School-Conference “Approximation and Data Analysis”, Nizhny Novgorod, Russia |
| 09/2019 | <b>A Guided Walk Through the ZOO of Stochastic Gradient Descent Methods</b> (Mini-course, 5 hours), Moscow Institute of Physics and Technology, Dolgoprudny, Russia                |

|         |                                                                                                                                                                                                                                                                                                     |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08/2019 | <b>A Guided Walk Through the ZOO of Stochastic Gradient Descent Methods</b> (Summer School Lectures, 6 hours), International Conference on Continuous Optimization (ICCOPT 2019), Berlin, Germany                                                                                                   |
| 02/2019 | <b>Randomized Optimization Methods</b> (PhD Course, 4.5 hours), Erwin Schrödinger International Institute for Mathematics and Physics (ESI), Vienna, Austria                                                                                                                                        |
| 06/2018 | <b>Stochastic Reformulations in Linear Algebra and Optimization</b> (Summer School, 2 hours), Control, Information and Optimization, Voronovo, Moscow Region, Russia                                                                                                                                |
| 04/2018 | <b>Introduction to Optimization for Machine Learning</b> (short outreach course for selected Saudi university students who previously participated in the Saudi National Mathematical Olympiad or IMO, 4.5 hours), KAUST, Thuwal, KSA                                                               |
| 08/2017 | <b>Randomized Optimization Methods</b> (Summer School, 5 hours), Data Science Summer School (DS <sup>3</sup> ), École Polytechnique, France. Other courses: Joshua Bengio (Montreal), Deep Learning; Pradeep Ravikumar (CMU), Graphical Models; Csaba Szepesvári (Alberta/Google DeepMind), Bandits |
| 10/2015 | <b>Randomized Methods for Big Data: From Linear Systems to Optimization</b> (Tutorial), IEEE International Conference on Data Science and Advanced Analytics, Paris, France                                                                                                                         |
| 2015    | <b>Randomized Algorithms for Big Data Optimization</b> (PhD Course, 18 hours), Graduate School in Systems, Optimization, Control and Networks – Université catholique de Louvain, Belgium                                                                                                           |
| 09/2015 | <b>Optimization in Machine Learning</b> (PhD Course, 8 hours), Machine Learning Thematic Trimester, Toulouse, France                                                                                                                                                                                |
| 07/2015 | <b>Modern Convex Optimization Methods for Large-Scale Empirical Risk Minimization</b> (Tutorial, 2 hours, joint with M. Schmidt), ICML 2015, Lille, France                                                                                                                                          |
| 06/2014 | <b>Randomized Coordinate Descent Methods</b> (PhD Course, 6 hours), Khronos-Persyval Days “High-Dimensional Learning and Optimization”, Grenoble, France                                                                                                                                            |
| 06/2014 | <b>Coordinate Descent Methods</b> (Lecture, 2 hours), NATCOR PhD Course on Convex Optimization, Edinburgh, UK                                                                                                                                                                                       |
| 02/2014 | <b>Gradient Methods for Big Data</b> (Tutorial, 3 hours), Big Data: Challenges and Applications, Imperial College London, UK                                                                                                                                                                        |

## 9.4 TALKS @ RESEARCH SEMINARS

|      |                                                                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | University of Vienna                                                                                                                                                                                                                                     |
| 2025 | Kempelen Institute of Intelligent Technologies, The Hong Kong Polytechnic University, Federated Learning One World Seminar (virtual), Bielefeld University, KAUST (3)                                                                                    |
| 2024 | Tsinghua University, Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS), Peking University, KAUST (2)                                                                                                                             |
| 2023 | CMOR Special Lecture @ Rice University, Qualcomm AI Seminar, Apple, Kempelen Institute of Intelligent Technologies, Slovak Academy of Sciences (2)                                                                                                       |
| 2022 | Machine Learning NeEDS Mathematical Optimization (virtual), Federated Learning One World Seminar, KAUST (3), Better AI Meetup Bratislava, Hong Kong Baptist University, One World Seminar Series on the Mathematics of Machine Learning (virtual)        |
| 2021 | University of Tartu (virtual), Portsmouth (virtual), Kempelen Institute for Intelligent Technologies, Comenius University, MBZUAI (virtual), All Russian Seminar on Optimization (virtual), Federated Learning One World Seminar (virtual; 2), KAUST (3) |
| 2020 | ESET, Optimization One World Seminar, Montréal MLOpt Seminar                                                                                                                                                                                             |
| 2019 | Huawei                                                                                                                                                                                                                                                   |
| 2018 | Bratislava, KAUST (2), Warwick, Edinburgh (2)                                                                                                                                                                                                            |
| 2017 | Imperial College London, KAUST, Plymouth, Cardiff                                                                                                                                                                                                        |
| 2016 | Cambridge, Edinburgh (3), Stanford (2), KAUST, The Alan Turing Institute, LSE, Southampton, Skoltech, Yandex                                                                                                                                             |
| 2015 | Louvain, Oxford, IST Austria, UC Davis, UC Berkeley, Edinburgh                                                                                                                                                                                           |

|      |                                                    |
|------|----------------------------------------------------|
| 2014 | Moscow, Paris, Hong Kong, Edinburgh (3)            |
| 2013 | UC Berkeley, Google, SAS Inc, Louvain, Edinburgh   |
| 2012 | Wisconsin, Cambridge, Glasgow, Cardiff, Bratislava |
| 2011 | Edinburgh, Oxford, London, Heriot-Watt, Louvain    |
| 2010 | Birmingham, Nottingham, Southampton                |
| 2009 | ETH Zürich, Linz, Louvain, Edinburgh (2)           |
| 2008 | Liège, Bratislava                                  |
| 2007 | Cornell (2), Louvain (2)                           |

## 10. TEACHING<sup>82</sup>

|           |             |                                                    |
|-----------|-------------|----------------------------------------------------|
| KAUST     | Fall 2026   | Stochastic Gradient Descent Methods (CS 331)       |
|           | Fall 2025   | Introduction to Machine Learning (DSA 008)         |
|           | Fall 2025   | Federated Learning (CS 332)                        |
|           | Fall 2025   | Stochastic Gradient Descent Methods (CS 331)       |
|           | Fall 2025   | CS Graduate Seminar (CS 398)                       |
|           | Summer 2025 | Introduction to Optimization (DSA 211)             |
|           | Spring 2025 | Stochastic Gradient Descent Methods* (CS 331)      |
|           | Spring 2024 | Stochastic Gradient Descent Methods* (CS 331)      |
|           | Spring 2023 | Federated Learning* (CS 332)                       |
|           | Fall 2022   | Stochastic Gradient Descent Methods* (CS 331)      |
|           | Spring 2022 | Federated Learning* (CS 332)                       |
|           | Fall 2021   | Stochastic Gradient Descent Methods* (CS 331)      |
|           | Spring 2021 | Federated Learning* (CS 332)                       |
|           | Fall 2020   | Stochastic Gradient Descent Methods* (CS 331)      |
|           | Spring 2020 | Federated Learning* (CS 390T)                      |
|           | Spring 2019 | Contemporary Topics in Machine Learning* (CS 394D) |
|           | Spring 2018 | Contemporary Topics in Machine Learning* (CS 394D) |
|           | Fall 2019   | Big Data Optimization* (CS 390FF)                  |
|           | Fall 2018   | Big Data Optimization* (CS 390FF)                  |
|           | Fall 2017   | Big Data Optimization* (CS 390FF)                  |
| Edinburgh | Spring 2017 | Modern Optimization Methods for Big Data Problems* |
|           | Spring 2016 | Modern Optimization Methods for Big Data Problems* |
|           | Fall 2012   | Discrete Programming and Game Theory*              |
|           | Fall 2011   | Discrete Programming and Game Theory*              |
|           | Fall 2011   | Discrete Programming and Game Theory*              |
|           | Spring 2015 | Optimization Methods in Finance*                   |
|           | Spring 2014 | Optimization Methods in Finance*                   |
|           | Spring 2013 | Optimization Methods in Finance*                   |
|           | Spring 2012 | Optimization Methods in Finance*                   |
|           | Spring 2011 | Optimization Methods in Finance*                   |
|           | Fall 2012   | Game Theory*                                       |
|           | Fall 2011   | Game Theory*                                       |
|           | Fall 2010   | Game Theory*                                       |
|           | Spring 2013 | Computing and Numerics                             |
|           | Fall 2010   | Dynamic & Integer Programming                      |
|           | Fall 2010   | Mathematics for Chemical Engineers                 |
| Louvain   | Spring 2009 | Nonlinear Optimization (with Yu. Nesterov)         |
| Cornell   | Spring 2006 | Optimization II/Nonlinear Optimization             |
|           | Summer 2005 | Engineering Probability and Statistics*            |

<sup>82</sup>I have proposed and developed from scratch courses marked with an asterisk. I was the instructor for all courses marked in bold. I was a TA (teaching assistant / tutor) for all other courses.

|          |             |                                                                    |
|----------|-------------|--------------------------------------------------------------------|
|          | Fall 2003   | Engineering Probability and Statistics                             |
|          | Summer 2003 | Engineering Probability and Statistics                             |
|          | Spring 2004 | Optimization II                                                    |
|          | Spring 2005 | Application of Game Theory and OR to IT                            |
|          | Spring 2005 | Topics in Linear Optimization                                      |
|          | Fall 2006   | Combinatorial Optimization (PhD course taught by David Williamson) |
| Comenius | Fall 1998   | Complex Analysis                                                   |

## 11. CONFERENCE, STREAM, WORKSHOP & SEMINAR ORGANIZATION<sup>83</sup>

|             |                                                                                                                                                                                                                                            |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05/2026     | Optimization for Learning Workshop, Lund, Sweden (scientific committee)                                                                                                                                                                    |
| 11/2025     | KAUST Workshop on Distributed Training in the Age of Large Language Models, Thuwal, Saudi Arabia (organizer)                                                                                                                               |
| 10/2025     | The 3rd IEEE International Conference on Federated Learning Technologies and Applications (FLTA 2025), Dubrovnik, Croatia (general chair)                                                                                                  |
| 06/2025     | The 4th Workshop on Federated Learning for Computer Vision, CVPR 2025 (co-organizer)                                                                                                                                                       |
| 07/2024     | International Symposium on Mathematical Programming (ISMP) (stream co-organizer with Lin Xiao and Simon Lacoste-Julien)                                                                                                                    |
| 02/2023     | Rising Stars in AI Symposium, KAUST, Thuwal, Saudi Arabia                                                                                                                                                                                  |
| 12/2022     | Federated Learning Workshop, NeurIPS                                                                                                                                                                                                       |
| 03/2022     | Rising Stars in AI Symposium, KAUST, Thuwal, Saudi Arabia                                                                                                                                                                                  |
| 05/2021     | SIAM Conference on Optimization, Virtual (member of the organizing committee)                                                                                                                                                              |
| 06/2020–now | <b>Federated Learning One World Seminar (FLOW)</b> <sup>84</sup> (founder and chair of the organizing committee)                                                                                                                           |
| 11/2019     | KAUST-Tsinghua-Industry Workshop on Advances in Artificial Intelligence, KAUST, Thuwal, Saudi Arabia                                                                                                                                       |
| 06/2019     | Sparse Approximation and Sampling, The Alan Turing Institute, London                                                                                                                                                                       |
| 04/2019     | A Short Course on Deep Learning and the Latest AI Algorithms, KAUST, Saudi Arabia. A 2-day course delivered by Xavier Bresson, NTU, Singapore                                                                                              |
| 07/2018     | International Symposium on Mathematical Programming, Bordeaux, France. Scientific Committee Member for stream 4a: “Machine Learning, Big Data, Cloud Computing, and Huge-Scale Optimization” (with A. d’Aspremont, O. Beaumont and S. Sra) |
| 02/2018     | Optimization and Big Data 2018, KAUST (co-organizer with M. Canini)                                                                                                                                                                        |
| 2017–now    | All Hands Meetings on Big Data Optimization, KAUST (a weekly group research seminar)                                                                                                                                                       |
| 09/2016     | IMA Numerical Linear Algebra and Optimization, Birmingham, UK (co-organizing 2 minisymposia)                                                                                                                                               |
| 12/2015     | Mathematical Perspectives on Big Data, a joint meeting of the London and Edinburgh mathematical societies, celebrating 150th anniversary of the former, Edinburgh                                                                          |
| 12/2015     | Theoretical and Computational Approaches to Large-Scale Inverse Problems, Edinburgh (Alan Turing Institute Scoping Workshop)                                                                                                               |
| 11/2015     | Distributed Machine Learning and Optimization, Edinburgh (Alan Turing Institute Scoping Workshop)                                                                                                                                          |
| 05/2015     | Optimization and Big Data 2015, Edinburgh (founder and co-organizer; with Z. Qu)                                                                                                                                                           |
| 01/2015     | International BASP Frontiers Workshop 2015, Villars-sur-Ollon, Switzerland                                                                                                                                                                 |
| 12/2014     | Workshop: Numerical Algorithms and Intelligent Software, Edinburgh                                                                                                                                                                         |
| 09/2014     | 2 minisymposia at 4th IMA Conf. on Numerical Lin. Alg. and Optimisation, Birmingham                                                                                                                                                        |
| 05/2014     | Coordinate Descent Methods Symposium at the SIAM Conference on Optimization, San Diego (24 speakers)                                                                                                                                       |

<sup>83</sup>I am excluding organized conference sessions.

<sup>84</sup><https://sites.google.com/view/one-world-seminar-series-flow/home>

|           |                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2014–2017 | All Hands Meetings on Big Data Optimization, University of Edinburgh (a weekly interdisciplinary research seminar attended by faculty, postdocs and PhD students from the Schools of Mathematics, Engineering and Informatics and Heriot-Watt University)                                                                                                                                |
| 07/2013   | Cluster Co-Chair, “Convex and Nonsmooth Optimization” at the International Conference on Continuous Optimization (ICCOPT), Lisbon, Portugal (with F. Glineur). We organized 23 invited sessions in the cluster (=70 speakers). ICCOPT is the premiere conference in continuous optimization, taking place once in 3 years. Our cluster was twice as large as the second largest cluster. |
| 05/2013   | Optimization and Big Data 2013, Edinburgh, 64 participants (founder and organizer)                                                                                                                                                                                                                                                                                                       |
| 05/2012   | Optimization and Big Data 2012, Edinburgh, 62 participants (founder and organizer)                                                                                                                                                                                                                                                                                                       |
| 07/2011   | 2 minisymposia at 3rd IMA Conf. on Numerical Linear Algebra and Optimisation, Birmingham                                                                                                                                                                                                                                                                                                 |
| 07/2011   | 2 minisymposia at 24th Biennial Conf. on Numerical Analysis, Glasgow                                                                                                                                                                                                                                                                                                                     |

## 12. COMMISSIONS OF TRUST

### 12.1 EXTERNAL ACTIVITIES

|           |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2027      | <b>Area Chair</b> , AAAI                                                                                                                                                                                                                                                                                                                                                                                          |
| 2026      | Program Committee Member, OPT 2026: Optimization for Machine Learning (NeurIPS 2026 Workshop)                                                                                                                                                                                                                                                                                                                     |
| 2026      | Program Committee Member, International Conference on Computational Optimization (ICOMP 2026)                                                                                                                                                                                                                                                                                                                     |
| 2026      | Reviewer, Swiss National Science Foundation (SNSF) career fellowship                                                                                                                                                                                                                                                                                                                                              |
| 2026      | External PhD Examiner for Amir Ali Farzin, Australian National University (advisors: Philipp Braun and Iman Shames)                                                                                                                                                                                                                                                                                               |
| 2026      | <b>Senior Area Chair</b> , ICML                                                                                                                                                                                                                                                                                                                                                                                   |
| 2026      | <b>Area Chair</b> , NeurIPS                                                                                                                                                                                                                                                                                                                                                                                       |
| 2026      | <b>Area Chair</b> , ICLR                                                                                                                                                                                                                                                                                                                                                                                          |
| 2025      | <b>Senior Area Chair</b> , NeurIPS                                                                                                                                                                                                                                                                                                                                                                                |
| 2025      | <b>Area Chair</b> , ICML                                                                                                                                                                                                                                                                                                                                                                                          |
| 2025      | <b>Area Chair</b> , ICLR                                                                                                                                                                                                                                                                                                                                                                                          |
| 2024      | <b>Distinguished Jury Member</b> , the AGBA GenAI Innovation Series for evaluating the innovations for 15th Edition of annual Aegis Graham Bell Awards, supported by Ministry of Electronics and Information Technology, Ministry of Education, Government of India, New Delhi and Department of Science & Technology, Government of India, Country Partner Australian Trade and Investment Commission (Austrade) |
| 2024–now  | <b>Action Editor</b> , Journal of Machine Learning Research (JMLR)                                                                                                                                                                                                                                                                                                                                                |
| 2024      | <b>Area Chair</b> , NeurIPS                                                                                                                                                                                                                                                                                                                                                                                       |
| 2024      | <b>Area Chair</b> , ICML                                                                                                                                                                                                                                                                                                                                                                                          |
| 2024      | <b>Area Chair</b> , ICLR                                                                                                                                                                                                                                                                                                                                                                                          |
| 2023–now  | <b>Associate Editor</b> , Numerische Mathematik                                                                                                                                                                                                                                                                                                                                                                   |
| 2023      | External PhD Examiner for Lie He, EPFL (advisor: Martin Jaggi)                                                                                                                                                                                                                                                                                                                                                    |
| 2023      | External PhD Examiner for Othmane Marfoq, Inria Sophia Antipolis (advisor: Giovanni Neglia)                                                                                                                                                                                                                                                                                                                       |
| 2023      | Invited to serve as Area Chair for COLT 2023 (declined)                                                                                                                                                                                                                                                                                                                                                           |
| 2023      | <b>Area Chair</b> , NeurIPS                                                                                                                                                                                                                                                                                                                                                                                       |
| 2023      | <b>Area Chair</b> , ICML                                                                                                                                                                                                                                                                                                                                                                                          |
| 2023      | <b>Area Chair</b> , ICLR                                                                                                                                                                                                                                                                                                                                                                                          |
| 2022–2024 | <b>Action Editor</b> , Transactions on Machine Learning Research (TMLR)                                                                                                                                                                                                                                                                                                                                           |
| 2022      | <b>Area Chair</b> , NeurIPS                                                                                                                                                                                                                                                                                                                                                                                       |
| 2022      | <b>Area Chair</b> , ICML                                                                                                                                                                                                                                                                                                                                                                                          |
| 2022      | <b>Area Chair</b> , ICLR                                                                                                                                                                                                                                                                                                                                                                                          |

|           |                                                                                                                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021      | Habilitation <sup>85</sup> Committee Member for Dr. Aurélien Bellet, Inria Lille - Nord Europe, France (other committee members: Francis Bach, Kamalika Chaudhuri and Catuscia Palamidessi) |
| 2021      | <b>Area Chair</b> , NeurIPS, virtual                                                                                                                                                        |
| 2021      | <b>Area Chair</b> , ICML, virtual                                                                                                                                                           |
| 2021–2022 | <b>Area Editor</b> <sup>86</sup> , Journal of Optimization Theory and Applications                                                                                                          |
| 2021      | Reviewer of Hi!Paris Fellowship applications in machine learning <sup>87</sup>                                                                                                              |
| 2021      | Associate Editor (declined invite), Journal of Artificial Intelligence and Machine Learning                                                                                                 |
| 2021–now  | Research Mentor, Kempelen Institute of Intelligent Technologies, Bratislava, Slovakia                                                                                                       |
| 2021      | <b>Senior Program Committee Member</b> , IJCAI, Montréal, Canada                                                                                                                            |
| 2021      | <b>Area Chair</b> , ICLR, Vienna, Austria                                                                                                                                                   |
| 2021      | Program Committee Member, RSEML 2021: Towards Robust, Secure and Efficient Machine Learning (AAAI 2021 Workshop), virtual                                                                   |
| 2020      | External PhD Examiner for Axel Böhm, University of Vienna (advisor: Radu Ioan Bot)                                                                                                          |
| 2020      | External PhD Examiner for Dmitry Grishchenko, Université Grenoble Alpes (advisors: Franck Iutzeler, Jérôme Malick, and Massih-Reza Amini)                                                   |
| 2020      | <b>Area Chair</b> , NeurIPS, Vancouver, Canada                                                                                                                                              |
| 2020      | <b>Expert Reviewer</b> , ICML, Vienna, Austria                                                                                                                                              |
| 2020      | Program Committee Member, ICML International Workshop on Federated Learning for User Privacy and Data Confidentiality                                                                       |
| 2020      | Evaluator & Reviewer, European Commission H2020 grants                                                                                                                                      |
| 2020      | Evaluator & Reviewer, European Commission ICT grants totaling 40+ million EUR                                                                                                               |
| 2020      | Program Committee Member, International Workshop on Federated Learning for User Privacy and Data Confidentiality (IJCAI-PRICAI <sup>88</sup> ), Yokohama, Japan                             |
| 2020      | <b>Senior Program Committee Member</b> , IJCAI-PRICAI, Yokohama, Japan                                                                                                                      |
| 2019      | Program Committee Member, NeurIPS, Vancouver, Canada                                                                                                                                        |
| 2019      | Program Committee Member, AISTATS, Naha, Okinawa, Japan                                                                                                                                     |
| 2019      | External PhD Examiner for Benjamin Dubois, École des Ponts, France (advisor: G. Obozinski)                                                                                                  |
| 2019–now  | <b>Handling Editor</b> , Journal of Nonsmooth Analysis and Optimization                                                                                                                     |
| 2019      | <b>Senior Program Committee Member</b> , IJCAI, Macao, China                                                                                                                                |
| 2019      | <b>Area Chair</b> , ICML, Long Beach, California                                                                                                                                            |
| 2018–now  | <b>Associate Editor</b> , Optimization Methods and Software                                                                                                                                 |
| 2018      | Reviewer, Carnegie Trust, UK                                                                                                                                                                |
| 2018      | Program Committee Member, NeurIPS, Montreal, Canada                                                                                                                                         |
| 2018      | Program Committee Member, ICML, Stockholm, Sweden                                                                                                                                           |
| 2018      | Program Committee Member, ICLR, Vancouver, Canada                                                                                                                                           |
| 2017      | Program Committee Member, NeurIPS, Long Beach, USA                                                                                                                                          |
| 2017      | Program Committee Member, AAAI, New Orleans, USA                                                                                                                                            |
| 2017      | Reviewer, ERC (European Research Council) Consolidator Grants                                                                                                                               |
| 2016      | Habilitation Examiner for Nicolas Couellan, Institut de Mathématiques de Toulouse, Université Paul Sabatier, France (other examiners: Jean-Baptiste Hiriart-Urruty (Toulouse))              |
| 2016      | External PhD Examiner for Igor Colin, Télécom ParisTech, France (other examiners: Alexandre D’Aspremont (ENS) and Mikael Johansson (KTH))                                                   |
| 2016      | <b>Guest Editor</b> , Journal of Computational Mathematics (co-editors: Xiaojun Chen, Yuhong Dai, and Yinyu Ye)                                                                             |
| 2016      | Reviewer, EPSRC Programme Grant Scheme                                                                                                                                                      |
| 2016      | External PhD Examiner for Hamid Reza Feyzmahdavian, Automatic Control Department, KTH Royal Institute of Technology, Sweden                                                                 |

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<sup>85</sup>Habilitation á diriger des recherches

<sup>86</sup>area: Optimization for Machine Learning

<sup>87</sup>Hi!Paris is a new interdisciplinary center for research and education on AI and Data Analytics for Science, Business and Society launched by HEC Paris and Institut polytechnique de Paris (IP Paris). See [www.hi-paris.fr](http://www.hi-paris.fr)

<sup>88</sup>International Joint Conference on Artificial Intelligence – Pacific Rim International Conference on Artificial Intelligence

|           |                                                                                                                                                                                                                            |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2016      | Program Committee Member, Symposium on Distributed Information Processing, Optimization, and Resource Management over Networks, IEEE Global Conference on Signal and Information Processing, Greater Washington, D.C., USA |
| 2016      | Program Committee Member, NeurIPS, Barcelona, Spain                                                                                                                                                                        |
| 2016      | Program Committee Member, ICML, New York, USA                                                                                                                                                                              |
| 2016      | Program Committee Member, International Conference on Internet of Things and Big Data, Rome, Italy                                                                                                                         |
| 2015      | Program Committee Member, AISTATS, San Diego, California                                                                                                                                                                   |
| 2015      | Program Committee Member, 13th EUROPT Workshop on Advances in Continuous Optimization, Edinburgh                                                                                                                           |
| 2015      | Program Committee Member, ICML, Lille, France                                                                                                                                                                              |
| 2015      | External DPhil Examiner for Sheng Fang, Mathematical Institute, University of Oxford, UK (internal examiner: Jared Tanner)                                                                                                 |
| 2015      | Lead, Alan Turing Institute PhD Programme in Data Science (responsible, on behalf of the University of Edinburgh, for the development of the PhD programme, starting in 2017)                                              |
| 2015      | Evaluator & Reviewer, EU Horizon 2020 grants totaling 36.2 million EUR                                                                                                                                                     |
| 2015      | Reviewer for Leverhulme Trust (2×)                                                                                                                                                                                         |
| 2015      | Reviewer for Isaac Newton Trust                                                                                                                                                                                            |
| 2014–2020 | <b>Associate Editor</b> , Optimization (Frontiers in Applied Mathematics and Statistics)                                                                                                                                   |
| 2014–2017 | Steering Committee (representing School of Mathematics), Centre for Doctoral Training in Data Science, University of Edinburgh (£5.03m grant from EPSRC)                                                                   |
| 2013–2017 | Member, EPSRC Peer Review College                                                                                                                                                                                          |
| 2013      | Evaluator & Reviewer, EU FP7 grants totaling 42.5 million EUR.                                                                                                                                                             |
| 2013      | Chief Editor (declined invite), Statistics, Optimization and Computing (SOIC)                                                                                                                                              |
| 2012–2014 | Steering Committee (representing University of Edinburgh), Numerical Algorithms and Intelligent Software (£5m grant from EPSRC)                                                                                            |
| 2011–2017 | Reviewer, EPSRC                                                                                                                                                                                                            |
| 2011–2016 | Faculty Advisor, SIAM Edinburgh Student Chapter                                                                                                                                                                            |

## 12.2 JOURNAL REVIEWING

Mathematical Programming, SIAM Journal on Optimization, SIAM Review, Foundations of Computational Mathematics, Journal of Machine Learning Research, Machine Learning, IEEE Signal Processing, Symposium on Theory of Computing, Computational Optimization and Applications, Optimization Methods and Software, SIAM Journal on Computing, European Journal of Operational Research, Central European Journal of Operational Research, Journal of Global Optimization.

## 12.4 SERVICE @ KAUST

|           |                                                                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 2025      | PhD Proposal Examiner for Olga Mashkova, Computer Science                                                                                  |
| 2024      | PhD Proposal Examiner for Eslam Abdelrahman, Computer Science                                                                              |
| 2022–2025 | <b>Member, SDAIA-KAUST Center of Excellence in Data Science and AI</b>                                                                     |
| 2022–2024 | <b>Founding Member, KAUST AI Initiative</b>                                                                                                |
| 2022–2024 | <b>Member, AI Initiative Faculty Search Committee</b>                                                                                      |
| 2022      | PhD Proposal Examiner for Fatimah Zohra, Computer Science                                                                                  |
| 2022      | MS Thesis Examiner for Fernando Zhapa Camacho, Computer Science                                                                            |
| 2021–2022 | <b>Member, AI Initiative Advisory Board</b>                                                                                                |
| 2021      | PhD Proposal Examiner for Han Shao, Computer Science                                                                                       |
| 2020–2021 | <b>Chair, Machine Learning Faculty Search Committee</b>                                                                                    |
| 2020      | PhD Thesis Examiner for Adel Bibi, Computer Science (other examiners: Yi Ma (Berkeley), Wolfgang Heidrich (KAUST), Bernard Ghanem (KAUST)) |
| 2019–2021 | <b>Member, AI Initiative Committee</b>                                                                                                     |

|           |                                                                                       |
|-----------|---------------------------------------------------------------------------------------|
| 2019–2021 | Faculty Sponsor, KAUST ACM Student Chapter                                            |
| 2019–2020 | <b>Chair, Machine Learning Faculty Search Committee</b>                               |
| 2019      | PhD Proposal Examiner for Adel Bibi, Computer Science                                 |
| 2019      | Member, Research Strategic Plan Working Group (representing CEMSE)                    |
| 2018–2019 | <b>Chair, Artificial Intelligence Committee</b> <sup>89</sup>                         |
| 2018–now  | <b>Co-Founder, The Machine Learning Hub</b> (with M. Canini, B. Ghanem and P. Kalnis) |
| 2018–2019 | CS Program Curriculum Committee Member                                                |
| 2018      | CS Faculty Search Committee Member, Machine Learning                                  |
| 2017      | PhD Proposal Examiner for Khalil Elkhail, Electrical Engineering                      |
| 2017–2019 | Elected Member of the Academic Council                                                |
| 2017–2018 | Faculty Search Committee, Statistics and Computer Science                             |
| 2017      | Directed Research Project Evaluation Panel                                            |

## 12.5 SERVICE @ EDINBURGH

|           |                                                                                                                                                                                                                                                                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2016      | Recruitment Panel, Chancellor’s Fellowships in “Mathematics of Data Science” and “Industrial Mathematics”                                                                                                                                                                                                                                               |
| 2016      | Internal PhD Examiner for Zhanxing Zhu, School of Informatics, University of Edinburgh (external examiner: Manfred Oppel (TU Berlin))                                                                                                                                                                                                                   |
| 2015      | PhD Admissions, Data Science                                                                                                                                                                                                                                                                                                                            |
| 2015      | Recruitment Panel, Lectureship in “Mathematics of Data Science”                                                                                                                                                                                                                                                                                         |
| 2014–2015 | Part of a small team at Edinburgh assisting with a bid for <b>The Alan Turing Institute (UK National Data Science and AI Institute)</b> and subsequently with organizational planning. The bid was successful and University of Edinburgh became one of 5 founding institutions of the Alan Turing Institute (with Oxford, Cambridge, UCL and Warwick). |
| 2013–2016 | PhD Admissions, OR & Optimization                                                                                                                                                                                                                                                                                                                       |
| 2009–2015 | Director of Studies and Personal Tutor                                                                                                                                                                                                                                                                                                                  |
| 2009–2015 | MSc Projects Coordinator, OR and Optimization Programme                                                                                                                                                                                                                                                                                                 |

## 13. PROFESSIONAL AFFILIATIONS

Association for Computing Machinery (ACM)  
 Society for Industrial and Applied Mathematics (SIAM)  
 Mathematical Optimization Society (MOS)  
 Edinburgh Mathematical Society (EMS)  
 Isaac Newton Institute for Mathematical Sciences (INIMS)  
 Institute for Operations Research and Management Science (INFORMS)  
 Foundations of Computational Mathematics (FoCM)  
 Slovak Mathematical Society (SMS)

## 14. INDUSTRY INVOLVEMENT

### 14.1 INDUSTRY INVOLVEMENT: SUMMARY

| company         | paper(s)   | comment |
|-----------------|------------|---------|
| Yandex Research | [249, 266] |         |
| Sony AI         | [240]      |         |
| Shanghai AI Lab | [214]      |         |

<sup>89</sup>I led a university-wide committee tasked by the President of KAUST to prepare a document mapping current AI activity at KAUST and suggesting a plan for building the AI initiative at KAUST in the next 5 years; we have written a 100+ page report.

|                          |               |                                      |
|--------------------------|---------------|--------------------------------------|
| JD Explore Academy       | [196]         |                                      |
| Intel                    | [95]          |                                      |
| Microsoft Research       | [79, 95, 193] |                                      |
| IBM Research             | [22, 78, 158] |                                      |
| Samsung AI               |               |                                      |
| Facebook                 | [83, 187]     |                                      |
| Amazon                   | [49, 151]     |                                      |
| Google                   | [51, 52, 168] | co-development of Federated Learning |
| Barefoot Networks        | [95]          |                                      |
| Baidu                    | [29]          |                                      |
| Western General Hospital | [11]          |                                      |

In the past I have had research discussions with SAS, Twitter, Arup, British Geological Survey, Confbuzz and Scottish Financial Risk Academy.

## 14.2 INDUSTRY INVOLVEMENT: FEDERATED LEARNING (with Google)

Standard machine learning approaches require centralizing the training data on one machine or in a data-center. For models trained from user interaction with mobile devices, a new approach was just released by Google, a result of collaboration between Google, Jakub Konečný and myself. The new approach is called “Federated Learning”; it is described in my papers [51, 52] and two additional papers by Google. Federated Learning enables mobile phones to collaboratively learn a shared prediction model while keeping all the training data on device, decoupling the ability to do machine learning from the need to store the data in the cloud. This goes beyond the use of local models that make predictions on mobile devices by bringing model training to the device as well. **The technology is now in use by around 1 billion Android devices.**

The CEO of Google, Sundar Pichai, [said](#):

“... we continue to set the pace in machine learning and AI research. We introduced a new technique for training deep neural networks on mobile devices called Federated Learning. This technique enables people to run a shared machine learning model, while keeping the underlying data stored locally on mobile phones.”

The new technology is described in a Google Research Blog, dated April 2017, to a lay audience. Selected media coverage: [Forbes](#), [The Verge](#), [Quartz](#), [TechRepublic](#), [ZDNet](#), [Computer Business Review](#), [Motherboard Vice](#), [Infoworld](#), [Venturebeat](#), [Engadget](#), [Tech Narratives](#), [GadgetHacks](#), [BGR](#), [AndroidAuthority](#), [AndroidHeadlines](#), [Tom’s Guide](#), [Digital Trends](#), [The Exponential View](#), [9to5google](#).

## 14.3. INDUSTRY INVOLVEMENT: YOUTUBE (with Google)

An excerpt from a support letter written to me by David J Harper, the Head of EMEA University Relations, Google Switzerland, for the purpose of a (successful) grant application:

“Google recognizes the contributions of Dr Richtárik’s research to the field of big data optimization. We have invited him to deliver a talk on his research on parallel and distributed coordinate descent methods in our internal Machine Learning seminar. The talk took place in Mountain View, California, in September 2013 and was televised via our teleconference facilities to Google offices around the globe. A variant of the algorithm<sup>90</sup> developed by Dr. Richtárik is in operation at Google in the YouTube recommendation engine.”

## 14.4 INDUSTRY INVOLVEMENT: LLM QUANTIZATION (with Yandex Research)

<sup>90</sup>A variant of the method developed in [18, 24].

HIGGS (Hadamard Incoherence with Gaussian MSE-optimal GridS), a data-free method for compressing large language models, was developed in collaboration between Yandex Research, MIT, ISTA and KAUST, and is described in our paper “Pushing the limits of large language model quantization via the linearity theorem” (NAACL 2025). HIGGS requires neither calibration data nor gradient-based optimization, and models can be quantized in minutes directly on a laptop or smartphone, with no significant loss in quality. The method has been applied to Llama 3.1 and 3.2, DeepSeek and Qwen models, and Yandex uses it to prototype and accelerate product development. Yandex announced the method in a [press release](#) on 11 April 2025.

## 15. PUBLICATIONS

### 15.1 CITATION METRICS<sup>91</sup>

According to [Google Scholar](#), my works attracted more than 30,000 citations, and my **h-index** is 72.

### 15.2 CONFERENCE/JOURNAL ABBREVIATIONS

|               |                                                                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NeurIPS       | Annual Conference on Neural Information Processing Systems<br>(a leading conference in machine learning and artificial intelligence research)                                           |
| ICML          | International Conference on Machine Learning<br>(a leading conference in machine learning and artificial intelligence research)                                                         |
| ICLR          | International Conference on Learning Representations<br>(a leading conference in machine learning and artificial intelligence research)                                                 |
| AISTATS       | International Conference on Artificial Intelligence and Statistics                                                                                                                      |
| ALT           | International Conference on Algorithmic Learning Theory                                                                                                                                 |
| AAAI          | Conference on Artificial Intelligence                                                                                                                                                   |
| UAI           | Uncertainty in Artificial Intelligence                                                                                                                                                  |
| ECAI          | European Conference on Artificial Intelligence                                                                                                                                          |
| NAACL         | Annual Conference of the Nations of the Americas Chapter of the ACL                                                                                                                     |
| MSML          | Mathematical and Scientific Machine Learning                                                                                                                                            |
| JMLR          | Journal of Machine Learning Research                                                                                                                                                    |
| TMLR          | Transactions on Machine Learning Research                                                                                                                                               |
| ECML PKDD     | European Conf. on Machine Learning and Principles & Practice of Knowledge Discovery in Databases                                                                                        |
| DistributedML | International Workshop on Distributed Machine Learning                                                                                                                                  |
| ICCV          | IEEE International Conference on Computer Vision                                                                                                                                        |
| VMV           | Vision, Modeling and Visualization                                                                                                                                                      |
| MLSP          | IEEE International Workshop on Machine Learning for Signal Processing                                                                                                                   |
| PROMS         | Springer Proceedings in Mathematics & Statistics                                                                                                                                        |
| ICASSP        | International Conference on Acoustics, Speech, and Signal Processing<br>(world’s largest and most comprehensive technical conference focused on signal processing and its applications) |
| GlobalSIP     | IEEE Global Conference on Signal and Information Processing                                                                                                                             |
| Allerton      | Annual Allerton Conference on Communication, Control, and Computing                                                                                                                     |
| SPARS         | Proceedings of Signal Processing with Adaptive Sparse Structured Representations                                                                                                        |
| WACV          | IEEE Winter Conference on Applications in Computer Vision                                                                                                                               |
| SPIE          | Proceedings of the Society of Photo-Optical Instrumentation Engineers                                                                                                                   |
| OR            | Operations Research Proceedings                                                                                                                                                         |
| SIGCOMM       | ACM’s Special Interest Group on Data Communications, specializing in the field of communication and computer networks                                                                   |
| SOSP          | Workshop on AI Systems at Symposium on Operating Systems Principles                                                                                                                     |
| NSDI          | USENIX Symposium on Networked Systems Design and Implementation                                                                                                                         |

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<sup>91</sup>These citations metric were extracted via Google Scholar in September 2025.

### 15.3 LIST OF PUBLICATIONS, PREPRINTS & TECHNICAL REPORTS

The papers are listed in reverse chronological order in terms of their appearance online. The **arXiv** identifier is mentioned for papers which are not yet published. Coauthors marked with *(r)*, *(p)*, *(d)*, *(m)* and *(i)* were my *(r)*esearch scientists, *(p)*ostdocs, *(d)*octoral students, *(m)*aster students and *(i)*nterns at the time of writing, respectively.

- (320) E. Shulgin<sup>(d)</sup>, J. K. H. Franke, D. von Rütte, T. H. Zhang, N. Ajroldi, K. Pöppel, B. Schölkopf, A. Klein, P. Richtárik, and A. Orvieto  
**Deriving hyperparameter scaling laws via modern optimization theory**  
*NeurIPS 2026*
- (319) I. Sokolov<sup>(d)</sup>, A. Tyurin, and P. Richtárik  
**Rennala-NSGD: asynchronous stochastic optimization beyond Euclidean geometry**  
*NeurIPS 2026*
- (318) E. Shulgin<sup>(d)</sup>, T. Gadaev, S. Khirirat<sup>(p)</sup>, and P. Richtárik  
**Understanding MARS: when scaling momentum provably helps**  
*ICML 2026*
- (317) E. Shulgin<sup>(d)</sup>, M. Awad, P. Richtárik, and E. Gorbunov  
**General analysis of LMO-based optimizers: beyond bounded variance**  
*ICML 2026*
- (316) I. Ilin<sup>(d)</sup>, P. Zmushko, and P. Richtárik  
**Super-Tuning: from activation-aware pruning to sparse fine-tuning**  
[arXiv:2607.09287](#)
- (315) L. Yuan, Y. Yang, D. Guo, P. Richtárik, and Z. Lin  
**SpecGradFilter: a spectral gradient filtering framework for taming federated heterogeneity**  
[arXiv:2607.04189](#)
- (314) X. Qian<sup>(r)</sup> and P. Richtárik  
**Convergence analysis of Muon-type methods with inexact LMO in the degenerate case**  
[arXiv:2606.21581](#)
- (313) I. Sokolov<sup>(d)</sup>, L. Condat<sup>(r)</sup>, and P. Richtárik  
**SILAGE: memory-efficient, full-gradient-free nonconvex optimization for nested finite sums**  
[arXiv:2606.15832](#)
- (312) I. Ilin<sup>(d)</sup> and P. Richtárik  
**Demystifying pipeline parallelism: first theory for PipeDream**  
[arXiv:2606.03498](#)
- (311) A. Sadiev<sup>(d)</sup>, L. Condat<sup>(r)</sup>, and P. Richtárik  
**A unified primal-dual recipe for accelerating three-operator splitting methods**  
[arXiv:2605.26985](#)
- (310) Y. Maziane<sup>(d)</sup>, A. Mahran<sup>(d)</sup>, A. Maranjyan<sup>(p)</sup>, and P. Richtárik  
**LOSCAR-SGD: local SGD with communication-computation overlap and delay-corrected**

**sparse model averaging**  
*NeurIPS 2026*

- (309) Y. Demidovich<sup>(p)</sup>, A. Chakraborty, G. Malinovsky<sup>(d)</sup>, A. Nedić, and P. Richtárik  
**Distance-aware Muon: adaptive step scaling for normalized optimization**  
arXiv:2605.18999
- (308) A. Sadiev<sup>(d)</sup>, A. Maranjyan<sup>(p)</sup>, I. Ilin<sup>(d)</sup>, and P. Richtárik  
**Ringmaster LMO: asynchronous linear minimization oracle momentum method**  
arXiv:2605.18174
- (307) A. Mahran<sup>(d)</sup>, A. Maranjyan<sup>(p)</sup>, and P. Richtárik  
**Rescaled asynchronous SGD: optimal distributed optimization under data and system heterogeneity**  
*NeurIPS 2026*
- (306) Z. Tovmasyan<sup>(m)</sup>, A. Maranjyan<sup>(p)</sup>, and P. Richtárik  
**Rennala MVR: improved time complexity for parallel stochastic optimization via momentum-based variance reduction**  
arXiv:2605.08871
- (305) P. Richtárik, K. Grunkowska<sup>(d)</sup>, and H. Li<sup>(d)</sup>  
**Local LMO: constrained gradient optimization via a local linear minimization oracle**  
arXiv:2605.08850
- (304) K. Grunkowska<sup>(d)</sup>, H. Li<sup>(d)</sup>, X. Qian<sup>(r)</sup>, and P. Richtárik  
**Broximal alignment for global non-convex optimization**  
arXiv:2604.13483
- (303) X. Qian<sup>(r)</sup>, A. Gaponov<sup>(m)</sup>, G. Malinovsky<sup>(d)</sup>, and P. Richtárik  
**Communication-efficient Gluon in federated learning**  
arXiv:2604.10689
- (302) L. Condat<sup>(r)</sup>, A. Sadiev<sup>(d)</sup>, and P. Richtárik  
**A Nesterov-accelerated primal-dual splitting algorithm for convex nonsmooth optimization**  
arXiv:2604.09245
- (301) H. Li<sup>(d)</sup>, K. Grunkowska<sup>(d)</sup>, and P. Richtárik  
**Stabilized proximal point method via trust region control**  
arXiv:2604.02943
- (300) R. Islamov, G. Malinovsky<sup>(d)</sup>, A. Gaponov<sup>(m)</sup>, A. Lucchi, P. Richtárik, and E. Gorbunov  
**Byzantine-robust and differentially private federated optimization under weaker assumptions**  
*UAI 2026*
- (299) L. Condat<sup>(r)</sup>, A. Maranjyan<sup>(d)</sup>, and P. Richtárik  
**BiCoLoR: Communication-efficient optimization with bidirectional compression and local training**  
arXiv:2601.12400

## Prepared in 2025

- (298) E. Shulgin<sup>(d)</sup>, G. Malinovsky<sup>(d)</sup>, S. Khirirat<sup>(p)</sup>, and P. Richtárik  
**First provable guarantees for practical private FL: beyond restrictive assumptions**  
arXiv:2512.21521
- (297) A. Fradin<sup>(i)</sup>, A. Sadiev<sup>(d)</sup>, L. Condat<sup>(r)</sup>, and P. Richtárik  
**Tight lower bounds and optimal algorithms for stochastic nonconvex optimization with heavy-tailed noise**  
*AISTATS 2026*
- (296) X. Qian, H. Rammal<sup>(i)</sup>, D. Kovalev, and P. Richtárik  
**Muon is provably faster with momentum variance reduction**  
arXiv:2512.16598
- (295) S. Khirirat<sup>(p)</sup>, A. Sadiev<sup>(d)</sup>, Y. Demidovich<sup>(p)</sup>, and P. Richtárik  
**Better LMO-based momentum methods with second-order information**  
arXiv:2512.13227
- (294) A. Sadiev<sup>(d)</sup>, Y. Demidovich<sup>(p)</sup>, I. Sokolov<sup>(d)</sup>, G. Malinovsky<sup>(d)</sup>, S. Khirirat<sup>(p)</sup>, and P. Richtárik  
**Improved convergence in parameter-agnostic error feedback through momentum**  
arXiv:2511.14501
- (293) E. Shulgin<sup>(d)</sup>, S. AlRashed, F. Orabona, and P. Richtárik  
**Beyond the ideal: Analyzing the inexact Muon update**  
*AISTATS 2026*
- (292) A. Sadiev<sup>(d)</sup>, P. Richtárik, and I. Fatkhullin  
**Second-order optimization under heavy-tailed noise: Hessian clipping and sample complexity limits**  
*NeurIPS 2025*
- (291) K. Gruntkowska<sup>(d)</sup>, Y. Maziane<sup>(d)</sup>, Z. Qu, and P. Richtárik  
**Drop-Muon: Update less, converge faster**  
arXiv:2510.02239
- (290) K. Gruntkowska<sup>(d)</sup> and P. Richtárik  
**Non-Euclidean broximal point method: a blueprint for geometry-aware optimization**  
arXiv:2510.00823
- (289) K. Gruntkowska<sup>(d)</sup>, A. Gaponov<sup>(m)</sup>, Z. Tovmasyan<sup>(m)</sup>, and P. Richtárik  
**Error feedback for Muon and friends**  
*ICLR 2026*
- (288) A. Fradin<sup>(i)</sup>, P. Richtárik and A. Tyurin  
**Local SGD and federated averaging through the lens of time complexity**  
arXiv:2509.23207
- (287) A. Maranjyan<sup>(d)</sup> and P. Richtárik  
**Ringleader ASGD: The first asynchronous SGD with optimal time complexity under data heterogeneity**  
*ICLR 2026*

- (286) L. Condat<sup>(r)</sup> and P. Richtárik  
**Convergence analysis of the PAGE stochastic algorithm for convex finite-sum optimization**  
*Journal of Optimization Theory and Applications* 210:33, 2026
- (285) I. Sokolov<sup>(d)</sup>, A. Sadiev<sup>(d)</sup>, Y. Demidovich<sup>(p)</sup>, F. S. Al-Qahtani, and P. Richtárik  
**Bernoulli-LoRA: A theoretical framework for randomized low-rank adaptation**  
arXiv:2508.03820
- (284) A. Riabinin<sup>(d)</sup>, E. Shulgin<sup>(d)</sup>, K. Gruntkowska<sup>(d)</sup>, P. Richtárik  
**Gluon: Making Muon & Scion great again! (Bridging theory and practice of LMO-based optimizers for LLMs)**  
arXiv:2505.13416
- (283) L. Condat<sup>(r)</sup>, E. Gasanov<sup>(d)</sup>, P. Richtárik  
**The stochastic multi-proximal method for nonsmooth optimization**  
arXiv:2505.12409
- (282) I. Ilin<sup>(d)</sup>, P. Richtárik  
**Thanos: A block-wise pruning algorithm for efficient large language model compression**  
arXiv:2504.05346
- (281) A. Beikmohammadi, S. Khirirat<sup>(p)</sup>, P. Richtárik, and S. Magnússon  
**Collaborative value function estimation under model mismatch: a federated temporal difference analysis**  
*ECML PKDD 2025*
- (280) K. Burlachenko<sup>(d)</sup>, P. Richtárik  
**BurTorch: Revisiting training from first principles by coupling autodiff, math optimization, and systems**  
arXiv:2503.13795
- (279) E. Shulgin<sup>(d)</sup>, S. Khirirat<sup>(p)</sup>, P. Richtárik  
**Smoothed normalization for efficient distributed private optimization**  
arXiv:2502.13482
- (278) A. Riabinin<sup>(d)</sup>, A. Khaled, P. Richtárik  
**A novel unified parametric assumption for nonconvex optimization**  
arXiv:2502.12329
- (277) R. Islamov, S. Horváth, A. Lucchi, P. Richtárik, E. Gorbunov  
**Double momentum and error feedback for clipping with fast rates and differential privacy**  
arXiv:2502.11682
- (276) Z. Tovmasyan<sup>(i)</sup>, G. Malinovsky<sup>(d)</sup>, L. Condat<sup>(r)</sup>, P. Richtárik  
**Revisiting stochastic proximal point methods: generalized smoothness and similarity**  
*Journal of Nonlinear and Variational Analysis* 10(3):471–505, 2026
- (275) K. Gruntkowska<sup>(d)</sup>, H. Li<sup>(d)</sup>, A. Rane<sup>(i)</sup>, P. Richtárik  
**The ball-proximal (=”broximal”) point method: a new algorithm, convergence theory, and applications**

arXiv:2502.02002

- (274) A. Maranjyan<sup>(d)</sup>, E.M. Saad, P. Richtárik, F. Orabona  
**ATA: Adaptive task allocation for efficient resource management in distributed machine learning**  
*ICML 2025*
- (273) K. Yi<sup>(d)</sup>, P. Richtárik  
**Symmetric pruning of large language models**  
arXiv:2501.18980
- (272) A. Maranjyan<sup>(d)</sup>, A. Tyurin, P. Richtárik  
**Ringmaster ASGD: The first asynchronous SGD with optimal time complexity**  
*ICML 2025*

## Prepared in 2024

- (271) E. Shulgin<sup>(d)</sup>, P. Richtárik  
**On the convergence of DP-SGD with adaptive clipping**  
arXiv:2412.19916
- (270) I. Sokolov<sup>(d)</sup>, P. Richtárik  
**MARINA-P: Superior performance in non-smooth federated optimization with adaptive stepsizes**  
arXiv:2412.17082
- (269) A. Maranjyan<sup>(d)</sup>, A. Sadiev<sup>(d)</sup>, P. Richtárik  
**Differentially private random block coordinate descent**  
arXiv:2412.17054
- (268) E. Gasanov<sup>(d)</sup>, P. Richtárik  
**Speeding up stochastic proximal optimization in the high Hessian dissimilarity setting**  
arXiv:2412.13619
- (267) Y. Demidovich<sup>(p)</sup>, P. Ostroukhov, G. Malinovsky<sup>(d)</sup>, S. Horváth, M. Takáč, P. Richtárik, E. Gorbunov  
**Methods with local steps and random reshuffling for generally smooth non-convex federated optimization**  
*ICLR 2025*
- (266) V. Malinovskii, A. Panferov, I. Ilin<sup>(m)</sup>, H. Guo, P. Richtárik, D. Alistarh  
**Pushing the limits of large language model quantization via the linearity theorem**  
*NAACL 2025*
- (265) S. Khirirat<sup>(p)</sup>, A. Sadiev<sup>(d)</sup>, A. Riabinin<sup>(d)</sup>, E. Gorbunov, P. Richtárik  
**Error feedback under  $(L_0, L_1)$ -smoothness: normalization and momentum**  
*NeurIPS 2025*
- (264) W. Anyszka<sup>(i)</sup>, K. Gruntkowska<sup>(d)</sup>, A. Tyurin<sup>(p)</sup>, P. Richtárik  
**Tighter performance theory of FedExProx**  
*ICLR 2026*

- (263) K. Burlachenko<sup>(d)</sup>, P. Richtárik  
**Unlocking FedNL: Self-contained compute-optimized implementation**  
arXiv:2410.08760
- (262) Grigory Malinovsky<sup>(d)</sup>, Umberto Michieli, Hasan Abed Al Kader Hammoud, Taha Ceritli, Hayder Elesedy, Mete Ozay, P. Richtárik  
**Randomized asymmetric chain of LoRA: The first meaningful theoretical framework for low-rank adaptation**  
arXiv:2410.08305
- (261) A. Maranjyan<sup>(d)</sup>, O. S. Omar<sup>(m)</sup>, P. Richtárik  
**MindFlayer: Efficient asynchronous parallel SGD in the presence of heterogeneous and random worker compute times**  
*UAI 2025*
- (260) H. Li<sup>(d)</sup>, P. Richtárik  
**On the convergence of FedProx with extrapolation and inexact prox**  
arXiv:2410.01410
- (259) E. Gorbunov, N. Tupitsa, S. Choudhury, A. Aliev, P. Richtárik, S. Horváth, M. Takáč  
**Methods for convex  $(L_0, L_1)$ -smooth optimization: clipping, acceleration, and adaptivity**  
*ICLR 2025*
- (258) K. Yi<sup>(d)</sup>, T. Kharisov<sup>(i)</sup>, I. Sokolov<sup>(d)</sup>, P. Richtárik  
**Cohort squeeze: Beyond a single communication round per cohort in cross-device federated learning**  
*NeurIPS 2024 FL Workshop*  
[Oral at the NeurIPS 2024 FL Workshop](#)
- (257) G. Meinhardt<sup>(i)</sup>, K. Yi<sup>(d)</sup>, L. Condat<sup>(r)</sup>, P. Richtárik  
**Prune at the clients, not the server: Accelerated sparse training in federated learning**  
arXiv:2405.20623
- (256) A. Karagulyan<sup>(p)</sup>, E. Shulgin<sup>(d)</sup>, A. Sadiev<sup>(d)</sup>, P. Richtárik  
**SPAM: Stochastic proximal point method with momentum variance reduction for non-convex cross-device federated learning**  
arXiv:2405.20127
- (255) L. Condat<sup>(r)</sup>, P. Richtárik  
**A simple linear convergence analysis of the Point-SAGA algorithm**  
arXiv:2405.19951
- (254) P. Richtárik, S. M. Giancola<sup>(i)</sup>, D. Lubczyk<sup>(i)</sup>, R. Yadav<sup>(i)</sup>  
**Local curvature descent: squeezing more curvature out of standard and Polyak gradient descent**  
*NeurIPS 2025*
- (253) A. Tyurin<sup>(p)</sup>, P. Richtárik  
**On the optimal time complexities in decentralized stochastic asynchronous optimization**  
*NeurIPS 2024*
- (252) P. Richtárik, A. Sadiev<sup>(d)</sup>, Y. Demidovich<sup>(p)</sup>  
**A unified theory of stochastic proximal point methods without smoothness**

arXiv:2405.15941

- (251) I.-V. Modoranu, M. Safaryan, G. Malinovsky<sup>(d)</sup>, E. Kurtic, T. Robert, P. Richtárik, D. Alistarh  
**MicroAdam: Accurate adaptive optimization with low space overhead and provable convergence**  
*NeurIPS 2024*
- (250) A. Tyurin<sup>(p)</sup>, K. Gruntkowska<sup>(d)</sup>, P. Richtárik  
**Freya PAGE: First optimal time complexity for large-scale nonconvex finite-sum optimization with heterogeneous asynchronous computations**  
*NeurIPS 2024*
- (249) V. Malinovskii, D. Mazur, I. Ilin<sup>(d)</sup>, D. Kuznedelev, K. Burlachenko<sup>(d)</sup>, K. Yi<sup>(d)</sup>, D. Alistarh, P. Richtárik  
**PV-Tuning: Beyond straight-through estimation for extreme LLM compression**  
*NeurIPS 2024*  
Oral at NeurIPS 2024 (0.4% acceptance rate)
- (248) A. Sadiev<sup>(d)</sup>, L. Condat<sup>(r)</sup>, P. Richtárik  
**Stochastic proximal point methods for monotone inclusions under expected similarity**  
arXiv:2405.14255
- (247) H. Li<sup>(d)</sup>, K. Acharya<sup>(i)</sup>, P. Richtárik  
**The power of extrapolation in federated learning**  
*NeurIPS 2024*
- (246) Kai Yi<sup>(d)</sup>, G. Meinhardt<sup>(i)</sup>, P. Richtárik  
**FedComLoc: Communication-efficient distributed training of sparse and quantized models**  
arXiv:2403.09904
- (245) Y. Demidovich<sup>(p)</sup>, G. Malinovsky<sup>(d)</sup>, P. Richtárik  
**Streamlining in the Riemannian realm: Efficient Riemannian optimization with loopless variance reduction**  
arXiv:2403.06677  
Best Paper Award (runner-up), International Conference on Computational Optimization (ICOMP 2025)
- (244) L. Condat<sup>(r)</sup>, A. Maranjyan<sup>(d)</sup>, P. Richtárik  
**LoCoDL: Communication-efficient distributed learning with local training and compression**  
*ICLR 2025*  
Spotlight at ICLR 2025
- (243) K. Gruntkowska<sup>(d)</sup>, A. Tyurin<sup>(p)</sup>, P. Richtárik  
**Improving the worst-case bidirectional communication complexity for nonconvex distributed optimization under function similarity**  
*NeurIPS 2024*  
Spotlight at NeurIPS 2024
- (242) A. Tyurin<sup>(p)</sup>, M. Pozzi<sup>(i)</sup>, I. Ilin<sup>(d)</sup>, P. Richtárik  
**Shadowheart SGD: distributed asynchronous SGD with optimal time complexity under arbitrary computation and communication heterogeneity**

- (241) A. Panferov<sup>(i)</sup>, Y. Demidovich<sup>(p)</sup>, A. Rammal<sup>(i)</sup>, P. Richtárik  
**Correlated quantization for faster nonconvex distributed optimization**  
*UAI 2025*  
Oral at UAI 2025

## Prepared in 2023

- (240) K. Yi<sup>(d)</sup>, N. Gazagnadou, P. Richtárik, and L. Lyu  
**FedP3: Personalized and privacy-friendly federated network pruning under model heterogeneity**  
*ICLR 2024*
- (239) P. Richtárik, E. Gasanov<sup>(d)</sup>, and K. Burlachenko<sup>(d)</sup>  
**Error feedback reloaded: From quadratic to arithmetic mean of smoothness constants**  
*ICLR 2024*
- (238) J. Xin, I. Ilin<sup>(d)</sup>, S. Zhang, M. Canini, and P. Richtárik  
**Kimad: Adaptive gradient compression with bandwidth awareness**  
*DistributedML 2023*
- (237) K. Burlachenko<sup>(d)</sup>, A. Alrowithi, F. A. Albalawi, and P. Richtárik  
**Federated learning is better with non-homomorphic encryption**  
*DistributedML 2023*
- (236) Y. Demidovich<sup>(p)</sup>, G. Malinovsky<sup>(d)</sup>, E. Shulgin<sup>(d)</sup>, and P. Richtárik  
**MAST: Model-agnostic sparsified training**  
*ICLR 2025*
- (235) G. Malinovsky<sup>(d)</sup>, P. Richtárik, S. Horváth, and E. Gorbunov  
**Byzantine robustness and partial participation can be achieved simultaneously: just clip gradient differences**  
arXiv:2311.14127
- (234) M. Fornasier, K. Riedl, P. Richtárik, and L. Sun<sup>(d)</sup>  
**Consensus-based optimization with truncated noise**  
*European Journal of Applied Mathematics, 2024*
- (233) A. Rammal<sup>(i)</sup>, K. Grutkowska<sup>(i)</sup>, N. Fedin, E. Gorbunov, and P. Richtárik  
**Communication compression for Byzantine robust learning: New efficient algorithms and improved rates**  
*AISTATS 2024*
- (232) H. Li<sup>(d)</sup>, A. Karagulyan<sup>(p)</sup>, and P. Richtárik  
**MARINA meets matrix stepsizes: Variance reduced distributed non-convex optimization**  
arXiv:2310.04614
- (231) E. Gorbunov, A. Sadiev<sup>(d)</sup>, M. Danilova, S. Horváth, G. Gidel, P. Dvurechensky, A. Gasnikov, and P. Richtárik  
**High-probability convergence for composite and distributed stochastic minimization**

- and variational inequalities with heavy-tailed noise**  
*ICML 2024*  
 Oral (144/9473 = top 1.5%)
- (230) E. Shulgin<sup>(d)</sup> and P. Richtárik  
**Towards a better theoretical understanding of independent subnetwork training**  
*ICML 2024*
- (229) R. Szlendak<sup>(m)</sup>, E. Gasanov<sup>(d)</sup>, and P. Richtárik  
**Understanding progressive training through the framework of randomized coordinate descent**  
*AISTATS 2024*
- (228) M. Grudzień, G. Malinovsky<sup>(d)</sup>, and P. Richtárik  
**Improving accelerated federated learning with compression and importance sampling**  
 arXiv:2306.03240
- (227) S. Khirirat<sup>(p)</sup>, E. Gorbunov, S. Horváth, R. Islamov, F. Karay, and P. Richtárik  
**Clip21: Error feedback for gradient clipping**  
 arXiv:2305.18929
- (226) J. Xin, M. Canini, P. Richtárik, and S. Horváth  
**Quantize once, train fast: allreduce-compatible compression with provable guarantees**  
*ECAI 2025*
- (225) Y. Demidovich<sup>(p)</sup>, G. Malinovsky<sup>(d)</sup>, I. Sokolov<sup>(d)</sup> and P. Richtárik  
**A guide through the zoo of biased SGD**  
*NeurIPS 2023*
- (224) P. Richtárik, E. Gasanov<sup>(d)</sup> and K. Burlachenko<sup>(d)</sup>  
**Error feedback shines when features are rare**  
 arXiv:2305.15264
- (223) I. Fatkhullin, A. Tyurin<sup>(p)</sup>, and P. Richtárik  
**Momentum provably improves error feedback!**  
*NeurIPS 2023*
- (222) K. Yi<sup>(d)</sup>, L. Condat<sup>(r)</sup>, and P. Richtárik  
**Explicit personalization and local training: double communication acceleration in federated learning**  
*TMLR 2025*
- (221) A. Tyurin<sup>(p)</sup> and P. Richtárik  
**Optimal time complexities of parallel stochastic optimization methods under a fixed computation model**  
*NeurIPS 2023*
- (220) A. Tyurin<sup>(p)</sup> and P. Richtárik  
**2Direction: Theoretically faster distributed training with bidirectional communication compression**  
*NeurIPS 2023*

- (219) H. Li<sup>(d)</sup>, A. Karagulyan<sup>(p)</sup> and P. Richtárik  
**Det-CGD: Compressed gradient descent with matrix stepsizes for non-convex optimization**  
*ICLR 2024*
- (218) A. Karagulyan<sup>(p)</sup> and P. Richtárik  
**ELF: Federated Langevin algorithms with primal, dual and bidirectional compression**  
*UAI 2025*
- (217) L. Condat<sup>(r)</sup>, G. Malinovsky<sup>(d)</sup>, and P. Richtárik  
**TAMUNA: Accelerated federated learning with local training and partial participation**  
*arXiv:2302.09832*
- (216) G. Malinovsky<sup>(d)</sup>, S. Horváth, K. Burlachenko<sup>(d)</sup> and P. Richtárik  
**Federated learning with regularized client participation**  
*arXiv:2302.03662*
- (215) A. Sadiev<sup>(d)</sup>, M. Danilova, E. Gorbunov, S. Horváth, G. Gidel, P. Dvurechensky, A. Gasnikov and P. Richtárik  
**High-probability bounds for stochastic optimization and variational inequalities: the case of unbounded variance**  
*ICML 2023*
- (214) X. Qian<sup>(p)</sup>, H. Dong, T. Zhang and P. Richtárik  
**Catalyst acceleration of error compensated methods leads to better communication complexity**  
*AISTATS 2023*
- (213) S. Hanzely<sup>(d)</sup>, K. Mishchenko<sup>(d)</sup> and P. Richtárik  
**Convergence of first-order algorithms for meta-learning with Moreau envelopes**  
*arXiv:2301.06806*

## Prepared in 2022

- (212) M. Grudziński<sup>(i)</sup>, G. Malinovsky<sup>(d)</sup> and P. Richtárik  
**Can 5th generation local training methods support client sampling? Yes!**  
*AISTATS 2023*
- (211) M. Makarenko, E. Gasanov<sup>(d)</sup>, R. Islamov<sup>(i)</sup>, A. Sadiev<sup>(d)</sup> and P. Richtárik  
**Adaptive compression for communication-efficient distributed training**  
*TMLR 2023*
- (210) S. Hanzely<sup>(d)</sup>, D. Kamzolov, D. Pasechnyuk, A. Gasnikov, P. Richtárik and M. Takáč  
**A damped Newton method achieves global  $O(1/k^2)$  and local quadratic convergence rate**  
*NeurIPS 2022*
- (209) A. Maranjyan<sup>(i)</sup>, M. Safaryan<sup>(p)</sup> and P. Richtárik  
**GradSkip: Communication-accelerated local gradient methods with better computational complexity**  
*TMLR 2025*

- (208) L. Condat<sup>(r)</sup>, I. Agarský<sup>(d)</sup> and P. Richtárik  
**CompressedScaffnew: the first theoretical double acceleration of communication from local training and compression in distributed optimization**  
*Optimization, 2026*
- (207) L. Sun<sup>(d)</sup> and P. Richtárik  
**Improved Stein variational gradient descent with importance weights**  
arXiv:2210.00462
- (206) K. Gruntkowska<sup>(i)</sup>, A. Tyurin<sup>(p)</sup> and P. Richtárik  
**EF21-P and friends: Improved theoretical communication complexity for distributed optimization with bidirectional compression**  
*ICML 2023*
- (205) S. Boucheroite, G. Malinovsky<sup>(d)</sup>, P. Richtárik and E. H. Bergou  
**Minibatch stochastic three points method for unconstrained smooth minimization**  
*AAAI 2024*
- (204) E. H. Bergou<sup>(r)</sup>, K. Burlachenko<sup>(d)</sup>, A. Dutta and P. Richtárik  
**Personalized federated learning with communication compression**  
*TMLR 2023*
- (203) S. Horváth<sup>(d)</sup>, K. Mishchenko<sup>(d)</sup> and P. Richtárik  
**Adaptive learning rates for faster stochastic gradient methods**  
arXiv:2208.05287
- (202) L. Condat<sup>(r)</sup> and P. Richtárik  
**RandProx: Primal-dual optimization algorithms with randomized proximal updates**  
*ICLR 2023*  
*OPT2022: 14th Annual Workshop on Opt. for Machine Learning (NeurIPS 2022 Workshop)*
- (201) G. Malinovsky<sup>(d)</sup>, K. Yi<sup>(d)</sup> and P. Richtárik  
**Variance reduced ProxSkip: Algorithm, theory and application to federated learning**  
*NeurIPS 2022*
- (200) A. Sadiev<sup>(i)</sup>, D. Kovalev<sup>(d)</sup> and P. Richtárik  
**Communication acceleration of local gradient methods via an accelerated primal-dual algorithm with inexact prox**  
*NeurIPS 2022*
- (199) E. Shulgin<sup>(d)</sup> and P. Richtárik  
**Shifted compression framework: generalizations and improvements**  
*UAI 2022*
- (198) L. Sun<sup>(d)</sup> and P. Richtárik  
**A note on the convergence of mirrored Stein variational gradient descent under  $(L_0, L_1)$  smoothness condition**  
arXiv:2206.09709
- (197) A. Sadiev<sup>(i)</sup>, G. Malinovsky<sup>(d)</sup>, E. Gorbunov, I. Sokolov<sup>(d)</sup>, A. Khaled, K. Burlachenko<sup>(d)</sup> and P. Richtárik  
**Don't compress gradients in random reshuffling: compress gradient differences**

- (196) R. Islamov<sup>(i)</sup>, X. Qian<sup>(p)</sup>, S. Hanzely<sup>(d)</sup>, M. Safaryan<sup>(p)</sup> and P. Richtárik  
**Distributed Newton-type methods with communication compression and Bernoulli aggregation**  
*TMLR 2023*  
*NeurIPS Workshop 2022 (Order up! The Benefits of Higher-Order Optimization in Machine Learning)*
- (195) M. Alfarra, J. C. Pérez, E. Shulgin<sup>(d)</sup>, P. Richtárik and B. Ghanem  
**Certified robustness in federated learning**  
*NeurIPS Workshop 2022 (Federated Learning)*
- (194) A. Tyurin<sup>(p)</sup>, L. Sun<sup>(d)</sup>, K. Burlachenko<sup>(d)</sup> and P. Richtárik  
**Sharper rates and flexible framework for nonconvex SGD with client and data sampling**  
*TMLR 2023*
- (193) L. Sun<sup>(d)</sup>, A. Salim and P. Richtárik  
**Federated sampling with Langevin algorithm under isoperimetry**  
*TMLR 2024*
- (192) E. Gorbunov, S. Horváth<sup>(d)</sup>, P. Richtárik and G. Gidel  
**Variance reduction is an antidote to Byzantines: better rates, weaker assumptions and communication compression as a cherry on the top**  
*ICLR 2023*
- (191) L. Sun<sup>(d)</sup>, A. Karagulyan<sup>(p)</sup> and P. Richtárik  
**Convergence of Stein variational gradient descent under a weaker smoothness condition**  
*AISTATS 2023*
- (190) A. Tyurin<sup>(p)</sup> and P. Richtárik  
**A computation and communication efficient method for distributed nonconvex problems in the partial participation setting**  
*NeurIPS 2023*
- (189) L. Condat<sup>(r)</sup>, K. Yi<sup>(d)</sup> and P. Richtárik  
**EF-BV: A unified theory of error feedback and variance reduction mechanisms for biased and unbiased compression in distributed optimization**  
*NeurIPS 2022*
- (188) G. Malinovsky<sup>(d)</sup> and P. Richtárik  
**Federated random reshuffling with compression and variance reduction**  
*arXiv:2205.03914*
- (187) S. Horváth<sup>(d)</sup>, M. Sanjabi, L. Xiao, P. Richtárik and M. Rabbat  
**FedShuffle: Recipes for better use of local work in federated learning**  
*TMLR 2022*
- (186) K. Mishchenko<sup>(d)</sup>, G. Malinovsky<sup>(d)</sup>, S. Stich and P. Richtárik  
**ProxSkip: Yes! Local gradient steps provably lead to communication acceleration! Finally!**  
*ICML 2022*

- (185) D. Kovalev<sup>(d)</sup>, A. Beznosikov, A. Sadiev, M. Pershianov, P. Richtárik and A. Gasnikov  
**Optimal algorithms for decentralized stochastic variational inequalities**  
*NeurIPS 2022*
- (184) A. Tyurin<sup>(p)</sup> and P. Richtárik  
**DASHA: Distributed nonconvex optimization with communication compression and optimal oracle complexity**  
*ICLR 2023*
- (183) P. Richtárik, I. Sokolov<sup>(m)</sup>, I. Fatkhullin<sup>(i)</sup>, E. Gasanov<sup>(d)</sup>, Z. Li<sup>(r)</sup> and E. Gorbunov  
**3PC: Three point compressors for communication-efficient distributed training and a better theory for lazy aggregation**  
*ICML 2022*
- (182) H. Zhao, B. Li, Z. Li<sup>(r)</sup>, P. Richtárik and Y. Chi  
**BEER: Fast  $O(1/T)$  rate for decentralized nonconvex optimization with communication compression**  
*NeurIPS 2022*
- (181) G. Malinovsky<sup>(d)</sup>, K. Mishchenko<sup>(d)</sup> and P. Richtárik  
**Server-side stepsizes and sampling without replacement provably help in federated optimization**  
*DistributedML 2023*

Prepared in 2021

- (180) D. Kovalev<sup>(d)</sup>, A. Gasnikov and P. Richtárik  
**Accelerated primal-dual gradient method for smooth and convex-concave saddle-point problems with bilinear coupling**  
*NeurIPS 2022*
- (179) H. Zhao, K. Burlachenko<sup>(d)</sup>, Z. Li<sup>(r)</sup> and Peter Richtárik  
**Faster rates for compressed federated learning with client-variance reduction**  
 To appear in: *SIAM Journal on Mathematics of Data Science*, 2023  
 arXiv:2112.13097
- (178) K. Burlachenko<sup>(d)</sup>, S. Horváth<sup>(d)</sup> and P. Richtárik  
**FL-PyTorch: Optimization research simulator for federated learning**  
*DistributedML 2021*
- (177) E. Gasanov<sup>(d)</sup>, A. Khaled, S. Horváth and P. Richtárik  
**FLIX: A simple and communication-efficient alternative to local methods in federated learning**  
*AISTATS 2022*
- (176) X. Qian<sup>(p)</sup>, R. Islamov<sup>(i)</sup>, M. Safaryan<sup>(p)</sup> and P. Richtárik  
**Basis matters: better communication-efficient second order methods for federated learning**  
*AISTATS 2022*

- (175) A. Beznosikov, P. Richtárik, M. Diskin, M. Ryabinin and A. Gasnikov  
**Distributed methods with compressed communication for solving variational inequalities, with theoretical guarantees**  
*NeurIPS 2022*
- (174) Rafał Szlendak<sup>(d)</sup>, A. Tyurin<sup>(p)</sup> and P. Richtárik  
**Permutation compressors for provably faster distributed nonconvex optimization**  
*ICLR 2022*
- (173) I. Fatkhullin<sup>(i)</sup>, I. Sokolov<sup>(d)</sup>, E. Gorbunov<sup>(d)</sup>, Z. Li<sup>(p)</sup> and P. Richtárik  
**EF21 with bells & whistles: practical algorithmic extensions of modern error feedback**  
*JMLR 26(189):1–50, 2025*
- (172) X. Qian<sup>(p)</sup>, H. Dong, P. Richtárik and T. Zhang  
**Error compensated loopless SVRG, Quartz, and SDCA for distributed optimization**  
*arXiv:2109.10049*
- (171) M. Jahani, S. Rusakov, Z. Shi, P. Richtárik, M. W. Mahoney and M. Takáč  
**Doubly adaptive scaled algorithm for machine learning using second-order information**  
*ICLR 2022*
- (170) H. Zhao, Z. Li<sup>(r)</sup> and P. Richtárik  
**FedPAGE: A fast local method for federated learning**  
*arXiv:2108.04755*
- (169) Z. Li<sup>(r)</sup> and P. Richtárik  
**CANITA: Faster rates for distributed convex optimization with communication compression**  
*NeurIPS 2021*
- (168) 50+ authors  
**A field guide to federated optimization**  
*arXiv:2107.06917*
- (167) P. Richtárik, I. Sokolov<sup>(m)</sup>, and I. Fatkhullin<sup>(i)</sup>  
**EF21: A new, simpler, theoretically better, and practically faster error feedback**  
*NeurIPS 2021*  
*NeurIPS 2021 oral paper (less than 1% acceptance rate)*
- (166) D. Kovalev<sup>(d)</sup>, E. Gasanov<sup>(d)</sup>, P. Richtárik, and A. Gasnikov  
**Lower bounds and optimal algorithms for smooth and strongly convex decentralized optimization over time-varying networks**  
*NeurIPS 2021*
- (165) B. Wang<sup>(i)</sup>, M. Safaryan<sup>(p)</sup>, and P. Richtárik  
**Theoretically better and numerically faster distributed optimization with smoothness-aware quantization techniques**  
*NeurIPS 2022*
- (164) A. Salim<sup>(p)</sup>, L. Sun<sup>(d)</sup>, and P. Richtárik  
**A convergence theory for SVGD in the population limit under Talagrand’s inequality**  
**T1**

- (163) L. Condat<sup>(r)</sup> and P. Richtárik  
**MURANA: A generic framework for stochastic variance-reduced optimization**  
*MSML 2022*
- (162) M. Safaryan<sup>(p)</sup>, R. Islamov<sup>(i)</sup>, X. Qian<sup>(p)</sup>, and P. Richtárik  
**FedNL: Making Newton-type methods applicable to federated learning**  
*ICML 2022*
- (161) G. Malinovsky<sup>(m)</sup>, A. Sailanbayev<sup>(d)</sup>, and P. Richtárik  
**Random reshuffling with variance reduction: new analysis and better rates**  
[arXiv:2104.09342](#)
- (160) Z. Li<sup>(r)</sup> and P. Richtárik  
**ZeroSARAH: Efficient nonconvex finite-sum optimization with zero full gradient computations**  
[arXiv:2103.01447](#)
- (159) A. Salim<sup>(p)</sup>, L. Condat<sup>(r)</sup>, D. Kovalev<sup>(d)</sup>, and P. Richtárik  
**An optimal algorithm for strongly convex minimization under affine constraints**  
*AISTATS 2022*
- (158) Z. Shi, N. Loizou, P. Richtárik, and M. Takáč  
**AI-SARAH: Adaptive and implicit stochastic recursive gradient methods**  
*TMLR 2023*
- (157) D. Kovalev<sup>(d)</sup>, E. Shulgin<sup>(m)</sup>, P. Richtárik, A. Rogozin<sup>(i)</sup>, and A. Gasnikov  
**ADOM: Accelerated decentralized optimization method for time-varying networks**  
*ICML 2021*
- (156) K. Mishchenko<sup>(d)</sup>, B. Wang<sup>(i)</sup>, D. Kovalev<sup>(d)</sup>, and P. Richtárik  
**IntSGD: Floatless compression of stochastic gradients**  
*ICLR 2022*  
[ICLR 2022 Spotlight paper](#)
- (155) M. Gorbunov<sup>(i)</sup>, K. Burlachenko<sup>(d)</sup>, Z. Li<sup>(r)</sup>, and P. Richtárik  
**MARINA: faster non-convex distributed learning with compression**  
*ICML 2021*
- (154) M. Safaryan<sup>(p)</sup>, F. Hanzely<sup>(d)</sup>, and P. Richtárik  
**Smoothness matrices beat smoothness constants: better communication compression techniques for distributed optimization**  
*NeurIPS 2021*
- (153) K. Islamov<sup>(i)</sup>, X. Qian<sup>(p)</sup>, and P. Richtárik  
**Distributed second order methods with fast rates and compressed communication**  
*ICML 2021*
- (152) K. Mishchenko<sup>(d)</sup>, A. Khaled<sup>(i)</sup>, and P. Richtárik  
**Proximal and federated random reshuffling**  
*ICML 2022*

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- (151) S. Horváth<sup>(d)</sup>, A. Klein, P. Richtárik, and C. Archambeau  
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- (150) X. Qian<sup>(p)</sup>, H. Dong, P. Richtárik, and T. Zhang  
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*OPT2020: 12th Annual Workshop on Optimization for Machine Learning (NeurIPS 2020 Workshop)*
- (149) X. Qian<sup>(p)</sup>, H. Dong, P. Richtárik, and T. Zhang  
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- (148) E. Gorbunov<sup>(i)</sup>, F. Hanzely<sup>(d)</sup>, and P. Richtárik  
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*AISTATS 2021*
- (147) D. Kovalev<sup>(d)</sup>, A. Koloskova, M. Jaggi, P. Richtárik, and S. U. Stich  
**A linearly convergent algorithm for decentralized optimization: sending less bits for free!**  
*AISTATS 2021*
- (146) W. Chen<sup>(i)</sup>, S. Horváth<sup>(d)</sup>, and P. Richtárik  
**Optimal client sampling for federated learning**  
*TMLR 2022*  
*Privacy Preserving Machine Learning (NeurIPS 2020 Workshop)*
- (145) E. Gorbunov<sup>(i)</sup>, D. Kovalev<sup>(d)</sup>, D. Makarenko, and P. Richtárik  
**Linearly converging error compensated SGD**  
*NeurIPS 2020*
- (144) A. Albasyoni<sup>(m)</sup>, M. Safaryan<sup>(p)</sup>, L. Condat<sup>(r)</sup>, and P. Richtárik  
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*SpicyFL 2020: NeurIPS Workshop on Scalability, Privacy, and Security in Federated Learning*
- (143) F. Hanzely<sup>(d)</sup>, S. Hanzely<sup>(m)</sup>, S. Horváth<sup>(d)</sup>, and P. Richtárik  
**Lower bounds and optimal algorithms for personalized federated learning**  
*NeurIPS 2020*
- (142) L. Condat<sup>(r)</sup>, G. Malinovsky<sup>(m)</sup>, and P. Richtárik  
**Distributed proximal splitting algorithms with rates and acceleration**  
*Frontiers in Signal Processing, section Signal Processing for Communications, 2022*  
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- (141) R. M. Gower, M. Schmidt, F. Bach, and P. Richtárik  
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*Proceedings of the IEEE 108(11):1968–1983, 2020*
- (140) X. Qian<sup>(p)</sup>, P. Richtárik, and T. Zhang  
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- (139) A. S. Berahas, M. Jahani, P. Richtárik, and M. Takáč  
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**PAGE: A simple and optimal probabilistic gradient estimator for nonconvex optimization**  
*ICML 2021*  
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Spotlight talk
- (137) D. Kovalev<sup>(d)</sup>, A. Salim<sup>(p)</sup>, and P. Richtárik  
**Optimal and practical algorithms for smooth and strongly convex decentralized optimization**  
*NeurIPS 2020*
- (136) Ahmed Khaled<sup>(i)</sup>, Othmane Sebbouh<sup>(i)</sup>, Nicolas Loizou, Robert M. Gower, and P. Richtárik  
**Unified analysis of stochastic gradient methods for composite convex and smooth optimization**  
arXiv:2006.11573
- (135) S. Horváth<sup>(d)</sup> and P. Richtárik  
**A better alternative to error feedback for communication-efficient distributed learning**  
*ICLR 2021*  
*SpicyFL 2020: NeurIPS Workshop on Scalability, Privacy, and Security in Federated Learning*  
The Best Paper Award at the NeurIPS 2020 Workshop on Scalability, Privacy, and Security in Federated Learning
- (134) A. Salim<sup>(p)</sup> and P. Richtárik  
**Primal dual interpretation of the proximal stochastic gradient Langevin algorithm**  
*NeurIPS 2020*
- (133) Z. Li<sup>(p)</sup> and P. Richtárik  
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- (132) K. Mishchenko<sup>(d)</sup>, A. Khaled<sup>(i)</sup>, and P. Richtárik  
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*NeurIPS 2020*
- (131) M. Alfarra<sup>(m)</sup>, S. Hanzely<sup>(m)</sup>, A. Albasyoni<sup>(m)</sup>, B. Ghanem, and P. Richtárik  
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- (130) A. Salim<sup>(p)</sup>, L. Condat<sup>(r)</sup>, K. Mishchenko<sup>(d)</sup>, and P. Richtárik  
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*Linear Algebra and its Applications* 663:1–31, 2023
- (128) G. Malinovsky<sup>(i)</sup>, D. Kovalev<sup>(d)</sup>, E. Gasanov<sup>(d)</sup>, L. Condat<sup>(r)</sup>, and P. Richtárik  
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*ICML 2020*
- (127) A. Beznosikov<sup>(i)</sup>, S. Horváth<sup>(d)</sup>, P. Richtárik and M. Safaryan<sup>(p)</sup>  
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*ICML 2020*
- (125) D. Kovalev<sup>(d)</sup>, R. M. Gower, P. Richtárik and A. Rogozin<sup>(i)</sup>  
**Fast linear convergence of randomized BFGS**  
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- (124) F. Hanzely<sup>(d)</sup>, N. Doikov, P. Richtárik and Yu. Nesterov  
**Stochastic subspace cubic Newton method**  
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- (123) Mher Safaryan<sup>(p)</sup>, Egor Shulgin<sup>(m)</sup> and P. Richtárik  
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**Tighter theory for local SGD on identical and heterogeneous data**

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- (116) S. Horváth<sup>(d)</sup>, C.-Y. Ho, Ľ. Horváth<sup>(i)</sup>, A. Narayan Sahu, M. Canini and P. Richtárik  
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*Workshop on AI Systems at Symposium on Operating Systems Principles 2019 (SOSP'19)*
- (115) D. Kovalev<sup>(m)</sup>, K. Mishchenko<sup>(d)</sup> and P. Richtárik  
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*NeurIPS 2019 Workshop: Beyond First Order Methods in ML*
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- (112) A. Khaled<sup>(i)</sup>, K. Mishchenko<sup>(d)</sup> and P. Richtárik  
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*NeurIPS 2019 Workshop: Federated Learning for Data Privacy and Confidentiality*
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*International Symposium on Vision, Modeling and Visualization, 2019*  
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- (110) X. Qian<sup>(p)</sup>, Z. Qu and P. Richtárik  
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- (107) M. Safaryan and P. Richtárik  
**Stochastic sign descent methods: New algorithms and better theory**  
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- (106) A. Salim<sup>(p)</sup>, D. Kovalev<sup>(m)</sup> and P. Richtárik  
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*NeurIPS 2019*

- (105) E. H. Bergou<sup>(p)</sup>, M. Canini, A. Dutta<sup>(p)</sup>, P. Richtárik and Y. Xiao<sup>(i)</sup>  
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- (104) K. Mishchenko<sup>(d)</sup> and P. Richtárik  
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*AISTATS 2020*  
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- (102) F. Hanzely<sup>(d)</sup> and P. Richtárik  
**One method to rule them all: variance reduction for data, parameters and many new methods**  
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- (101) E. Gorbunov<sup>(i)</sup>, F. Hanzely<sup>(d)</sup> and P. Richtárik  
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- (100) S. Horváth<sup>(d)</sup>, C.Y. Ho, L. Horváth<sup>(i)</sup>, A. N. Sahu, M. Canini and P. Richtárik  
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- (99) R. M. Gower, D. Kovalev<sup>(m)</sup>, F. Lieder and P. Richtárik  
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**Scaling distributed machine learning with in-network aggregation**  
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- (94) S. Horváth<sup>(d)</sup>, D. Kovalev<sup>(d)</sup>, K. Mishchenko<sup>(d)</sup>, P. Richtárik and S. Stich  
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- (93) E. H. Bergou<sup>(p)</sup>, E. Gorbunov<sup>(i)</sup> and P. Richtárik  
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- (92) E. H. Bergou<sup>(p)</sup>, A. Bibi, B. Ghanem, O. Sener and P. Richtárik  
**A stochastic derivative-free optimization method with importance sampling**  
*AAAI* 2020
- (91) K. Mishchenko<sup>(d)</sup>, F. Hanzely<sup>(d)</sup> and P. Richtárik  
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- (81) S. Horváth<sup>(d)</sup> and P. Richtárik  
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- (80) F. Hanzely<sup>(d)</sup>, K. Mishchenko<sup>(d)</sup> and P. Richtárik  
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- (60) M. J. Ehrhardt, P. Markiewicz, A. Chambolle, P. Richtárik, J. Schott and C. B. Schönlieb  
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**Even faster accelerated coordinate descent using non-uniform sampling**  
*ICML 2016*
- (42) R. M. Gower<sup>(d)</sup> and P. Richtárik  
**Stochastic dual ascent for solving linear systems**  
arXiv:1512.06890
- (41) C. Ma, J. Konečný<sup>(d)</sup>, M. Jaggi, V. Smith, M. I. Jordan, P. Richtárik and M. Takáč  
**Distributed optimization with arbitrary local solvers**  
*Optimization Methods and Software 32(4):813–848, 2017*  
1st Most-Read Paper in Optimization Methods and Software, 2017
- (40) M. Takáč, P. Richtárik and N. Srebro  
**Distributed minibatch SDCA**  
To appear in: *JMLR*<sup>95</sup>
- (39) R. M. Gower<sup>(d)</sup> and P. Richtárik  
**Randomized iterative methods for linear systems**  
*SIAM Journal on Matrix Analysis and Applications 36(4):1660–1690, 2015*

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<sup>94</sup>A (refreshed) reprint of [21] originally published in SIAM Journal on Optimization

<sup>95</sup>We did not receive any reviews after 2.5 years since submission. The paper was recently accepted after a change in JMLR leadership.

Gower: 18th Leslie Fox Prize (2nd Prize), Institute for Mathematics and its Applications, 2017  
1st Most Downloaded Paper from the SIMAX Website, 2017  
2nd Most Downloaded Paper from the SIMAX Website, 2018  
2nd Most Downloaded Paper from the SIMAX Website, 2019  
1st Most Downloaded Paper from the SIMAX Website, 2020

- (38) D. Csiba<sup>(d)</sup> and P. Richtárik  
**Primal method for ERM with flexible mini-batching schemes and non-convex losses**  
arXiv:1506:02227
- (37) J. Konečný<sup>(d)</sup>, J. Liu, P. Richtárik and M. Takáč  
**Mini-batch semi-stochastic gradient descent in the proximal setting**  
*IEEE Journal of Selected Topics in Signal Processing* 10(2):242–255, 2016  
Konečný: BASP Frontiers Best Contribution Award, 2015
- (36) R. Tappenden<sup>(p)</sup>, M. Takáč<sup>(d)</sup> and P. Richtárik  
**On the complexity of parallel coordinate descent**  
*Optimization Methods and Software* 33(2), 372–395, 2018
- (35) D. Csiba<sup>(d)</sup>, Z. Qu<sup>(p)</sup> and P. Richtárik  
**Stochastic dual coordinate ascent with adaptive probabilities**  
*ICML 2015*  
Csiba: Best Contribution Award (2nd Prize), Optimization and Big Data 2015  
Implemented in Tensor Flow
- (34) C. Ma, V. Smith, M. Jaggi, M. I. Jordan, P. Richtárik and M. Takáč  
**Adding vs. averaging in distributed primal-dual optimization**  
*ICML 2015*  
Smith: MLconf Industry Impact Student Research Award, 2015  
CoCoA+ is now the default linear optimizer in Tensor Flow
- (33) Z. Qu<sup>(p)</sup>, P. Richtárik, M. Takáč<sup>(d)</sup> and O. Fercoq<sup>(p)</sup>  
**SDNA: Stochastic dual Newton ascent for empirical risk minimization**  
*ICML 2016*

## Prepared in 2014

- (32) Z. Qu<sup>(p)</sup> and P. Richtárik  
**Coordinate descent with arbitrary sampling II: expected separable overapproximation**  
*Optimization Methods and Software* 31(5):858–884, 2016  
7th Most-Read Paper in Optimization Methods and Software, 2017
- (31) Z. Qu<sup>(p)</sup> and P. Richtárik  
**Coordinate descent with arbitrary sampling I: algorithms and complexity**  
*Optimization Methods and Software* 31(5):829–857, 2016  
4th Most-Read Paper in Optimization Methods and Software, 2017
- (30) J. Konečný<sup>(d)</sup>, Z. Qu<sup>(p)</sup> and P. Richtárik  
**Semi-stochastic coordinate descent**  
*Optimization Methods and Software* 32(5):993–1005, 2017  
3rd Most-Read Paper in Optimization Methods and Software, 2017

- (29) Z. Qu<sup>(p)</sup>, P. Richtárik and T. Zhang  
**Quartz: Randomized dual coordinate ascent with arbitrary sampling**  
*NeurIPS 2015*
- (28) J. Konečný<sup>(d)</sup>, J. Liu, P. Richtárik and M. Takáč<sup>(d)</sup>  
**mS2GD: Mini-batch semi-stochastic gradient descent in the proximal setting**<sup>96</sup>  
*NeurIPS 2014 Workshop: Optimization for Machine Learning*
- (27) J. Konečný<sup>(d)</sup>, Z. Qu<sup>(p)</sup> and P. Richtárik  
**S2CD: Semi-stochastic coordinate descent**<sup>97</sup>  
*NeurIPS 2014 Workshop: Optimization for Machine Learning*
- (26) J. Konečný<sup>(d)</sup> and P. Richtárik  
**Simple complexity analysis of simplified direct search**  
 arXiv:1410.0390
- (25) J. Mareček<sup>(p)</sup>, P. Richtárik and M. Takáč<sup>(d)</sup>  
**Distributed block coordinate descent for minimizing partially separable functions**  
*PROMS 2015 (In: Al-Baali M., Grandinetti L., Purnama A. (eds) Numerical Analysis and Optimization. Springer Proceedings in Math. & Statistics, vol 134. Springer, Cham, 261–288, 2015)*
- (24) O. Fercoq<sup>(p)</sup>, Z. Qu<sup>(p)</sup>, P. Richtárik and M. Takáč<sup>(d)</sup>  
**Fast distributed coordinate descent for minimizing non-strongly convex losses**  
*MLSP 2014 (2014 IEEE Int. Workshop on Machine Learning for Signal Processing, 1–6, 2014)*
- (23) D. Forgan and P. Richtárik  
**On optimal solutions to planetesimal growth models**  
*Technical Report ERGO 14-002, 2014*
- (22) J. Mareček<sup>(p)</sup>, P. Richtárik and M. Takáč<sup>(d)</sup>  
**Matrix completion under interval uncertainty**  
*European Journal of Operational Research 256(1):35–43, 2017*

## Prepared in 2013

- (21) O. Fercoq<sup>(p)</sup> and P. Richtárik  
**Accelerated, parallel and proximal coordinate descent**  
*SIAM Journal on Optimization 25(4):1997–2023, 2015*  
 Fercoq: 17th Leslie Fox Prize (2nd Prize), Institute for Mathematics and its Applications, 2015  
 2nd Most Downloaded Paper from the SIOPT Website, 2016 & 2017
- (20) J. Konečný<sup>(d)</sup> and P. Richtárik  
**Semi-stochastic gradient descent**  
*Frontiers in Applied Mathematics and Statistics 3:9, 2017*
- (19) P. Richtárik and M. Takáč<sup>(d)</sup>  
**On optimal probabilities in stochastic coordinate descent methods**  
*Optimization Letters 10(6):1233–1243, 2016*

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<sup>96</sup>A short version of the journal paper [37]

<sup>97</sup>A short version of the journal paper [30]

- (18) P. Richtárik and M. Takáč<sup>(d)</sup>  
**Distributed coordinate descent method for learning with big data**  
*JMLR 17 (75):1–25, 2016*
- (17) O. Fercoq<sup>(p)</sup> and P. Richtárik  
**Smooth minimization of nonsmooth functions with parallel coordinate descent methods**  
*PROMS 2017 (Modelling and Optimization: Theory and Applications, Springer Proceedings in Math. and Statistics)*
- (16) R. Tappenden<sup>(p)</sup>, P. Richtárik and B. Büke  
**Separable approximations and decomposition methods for the augmented Lagrangian**  
*Optimization Methods and Software 30(3):643–668, 2015*
- (15) R. Tappenden<sup>(p)</sup>, P. Richtárik and J. Gondzio  
**Inexact coordinate descent: complexity and preconditioning**  
*Journal of Optimization Theory and Applications 171 (1):144–176, 2016*
- (14) M. Takáč<sup>(d)</sup>, S. D. Ahipasaoglu, N. M. Cheung and P. Richtárik  
**TOP-SPIN: TOPic discovery via Sparse Principal component INterference**  
*PROMS 2017 (Modelling and Optimization: Theory and Applications, Springer Proceedings in Math. and Statistics)*
- (13) M. Takáč<sup>(d)</sup>, A. Bijral, P. Richtárik and N. Srebro  
**Mini-batch primal and dual methods for SVMs**  
*ICML 2013*

## Prepared in 2012 or Before

- (12) P. Richtárik, M. Takáč<sup>(d)</sup>, S. D. Ahipasaoglu and M. Jahani  
**Alternating maximization: unifying framework for 8 sparse PCA formulations and efficient parallel codes**  
*Optimization and Engineering, 2020*
- (11) W. Hulme<sup>(m)</sup>, P. Richtárik, L. McGuire and A. Green  
**Optimal diagnostic tests for sporadic Creutzfeldt-Jakob disease based on SVM classification of RT-QuIC data**  
*Technical Report ERGO 12-014, 2012*  
[arXiv:1212.2617](#)
- (10) P. Richtárik and M. Takáč<sup>(d)</sup>  
**Parallel coordinate descent methods for big data optimization**  
*Mathematical Programming 156(1):433–484, 2016*  
[Takáč: 16th Leslie Fox Prize \(2nd Prize\), Institute for Mathematics and its Applications, 2013](#)
- (9) P. Richtárik and M. Takáč<sup>(d)</sup>  
**Efficient serial and parallel coordinate descent methods for huge-scale truss topology design**  
*In: Klatte D., Lüthi HJ., Schmedders K. (eds) Operations Research Proceedings 2011 (Gesellschaft für Operations Research e.V.). Springer, Berlin, Heidelberg, 2012*
- (8) P. Richtárik and M. Takáč<sup>(d)</sup>  
**Iteration complexity of randomized block-coordinate descent methods for minimizing**

**a composite function**

*Mathematical Programming* 144(2):1–38, 2014

Takáč: Best Student Paper Award (sole runner-up), INFORMS Computing Society, 2012

- (7) P. Richtárik and M. Takáč<sup>(d)</sup>  
**Efficiency of randomized coordinate descent methods on minimization problems with a composite objective function**  
*SPARS 2011 (Signal Processing with Adaptive Sparse Structured Representations)*
- (6) P. Richtárik  
**Finding sparse approximations to extreme eigenvectors: generalized power method for sparse PCA and extensions**  
*SPARS 2011 (Signal Processing with Adaptive Sparse Structured Representations)*
- (5) P. Richtárik  
**Approximate level method for nonsmooth convex optimization**  
*Journal of Optimization Theory and Applications* 152(2):334–350, 2012
- (4) M. Journée, Yu. Nesterov, P. Richtárik and R. Sepulchre  
**Generalized power method for sparse principal component analysis**  
*JMLR* 11:517–553, 2010
- (3) P. Richtárik  
**Improved algorithms for convex minimization in relative scale**  
*SIAM Journal on Optimization* 21(3):1141–1167, 2011
- (2) P. Richtárik  
**Simultaneously solving seven optimization problems in relative scale**  
*Technical Report, Optimization Online*, 2008
- (1) P. Richtárik  
**Some algorithms for large-scale linear and convex minimization in relative scale**  
*PhD thesis, School of ORIE, Cornell University*, 2007

## 15.5 PATENTS

- 2015      M. Takáč, S. D. Ahipasaoglu, P. Richtárik and N. M. Cheung  
            **Method and system for classifying images**  
            *Patent# WO/2015/011470*